PHYSICS

1) A boat is moving towards east with velocity 4m/s with respect to river water and river is flowing towards north with velocity 2 m/s. The wind is blowing towards north with velocity 6 m/s. The direction of waving of the flag hoisted on the boat, will be :-

(1) North-west

(2) South-East

(3) $\tan^{-1}(1/2)$ with east

(4) North

2) A particle is projected from horizontal making an angle 60° with initial velocity 40 m/sec. The approximate time taken by the particle to make angle 45° from horizontal, is :

(1) 15 sec.

(2) 2.0 sec.

(3) 20 sec.

(4) 1.5 sec.

3) The value of $\frac{3.91 \times 0.5142}{0.070}$ upto appropriate significant figures is

(1) 28.7217

(2) 28.72

(3) 28.7

(4) 29

4) A stone is rotated in a vertical circle with the help of a string. Speed at bottommost point is $\sqrt{8gR}$, where R is the radius of circle. The ratio of tension at the top and the bottom is :-

(1) 1 : 2

(2) 1 : 3

(3) 2 : 3

(4) 1 : 4

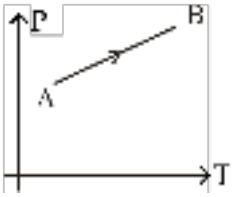
5) Two 50 gm ice cubes are dropped into 250 gm of water into a glass. If the water was initially at a temperature of 25°C and the temperature of ice –15°C. Then the final amount of water in the mixture is :-

(1) $\frac{1275}{4}$gm

(2) $\frac{125}{4}$gm

(3) $\frac{500}{4}$gm

(4) $\frac{800}{3}$gm

6) The (P – T) graph for given mass of an ideal gas is shown :- 

(1) $V_A > V_B$

(2) $V_A < V_B$

(3) $V_A = V_B$

(4) None of these

7) A sphere and a cube of same material and same volume are heated upto same temperature and allowed to cool in the same surroundings. The ratio of the amounts of radiations emitted will be:-

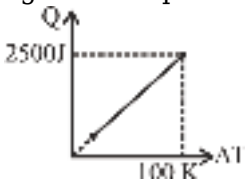
(1) 1 : 1

(2) $\frac{4\pi}{3} : 1$

(3) $\left(\frac{\pi}{6}\right)^{1/3} : 1$

(4) $\frac{1}{2}\left(\frac{4\pi}{3}\right)^{2/3} : 1$

8) One mole of a gas mixture is heated under constant pressure, and heat supplied Q is plotted against temperature difference acquired. Find the approximate value of γ for mixture :-



(1) $\frac{4}{3}$

(2) $\frac{3}{2}$

(3) $\frac{7}{5}$

(4) $\frac{5}{3}$

9) In equation $y = A\sin\left(\omega t + \frac{\pi}{4}\right)$, match the following for $y = \frac{A}{2}$:-

	Column-I		Column-II
(A)	Kinetic energy	(P)	Half the maximum value

(B)	Potential energy	(Q)	3/4 times the maximum value
(C)	Acceleration	(R)	1/4 times the maximum value
		(S)	Equal the maximum value

(1) A → P, B → S, C → R

(2) A → R, B → Q, C → P

(3) A → Q, B → S, C → P

(4) A → Q, B → R, C → P

10) An oil drop has a charge -9.6×10^{-19} C and has a mass 1.6×10^{-15}gm. When allowed to fall, due to air resistance it attains a constant velocity. Then if a uniform electric field is to be applied vertically to make the oil drop ascend up with the same constant speed, which of the following is correct :-
($g = 10\ m/s^2$)

(1) the electric field intensity is $\frac{1}{6} \times 10^5 N/C$ and directed upward

(2) the electric field intensity is $\frac{1}{6} \times 10^5 N/C$ and directed downward

(3) the intensity of the electric field is $\frac{1}{3} \times 10^2 N/C$ and directed downward

(4) the intensity of the electric field is $\frac{1}{3} \times 10^2 N/C$ and directed upward

11) A charge Q is distributed over three concentric spherical shells of radii a, b, c ($a < b < c$) such that their surface charge densities are equal to one another. The total potential at a point at distance r from their common centre, where $r < a$, would be :

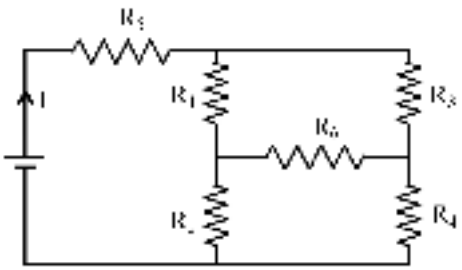
(1) $\frac{Q}{4\pi\varepsilon_0(a+b+c)}$

(2) $\frac{Q(a+b+c)}{4\pi\varepsilon_0(a^2+b^2+c^2)}$

(3) $\frac{Q}{12\pi\varepsilon_0}\frac{ab+bc+ca}{abc}$

(4) $\frac{Q}{4\pi\varepsilon_0}\frac{(a^2+b^2+c^2)}{(a^3+b^3+c^3)}$

12) In the given circuit, it is observed that the current I is independent of the value of the resistance



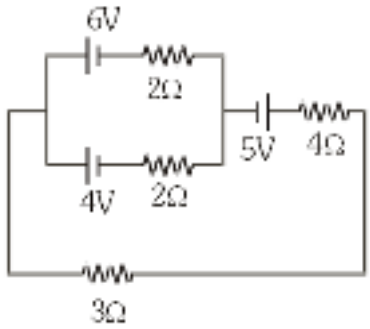
R_6. Then the resistance values must satisfy-

(1) $R_1R_2R_5 = R_3R_4R_6$

(2) $\frac{1}{R_5}+\frac{1}{R_6}=\frac{1}{R_1+R_2}+\frac{1}{R_3+R_4}$

(3) $R_1R_4 = R_2R_3$

(4) $R_1R_3 = R_2R_4 = R_5R_6$



13)
Calculate current in 5V battery :-

(1) $\frac{1}{4}$A

(2) $\frac{1}{8}$A

(3) $\frac{1}{2}$A

(4) Zero

14) Two resistances of 100 Ω and 200 Ω are connected in series with a battery of emf 4 volt and negligible internal resistance. A voltmeter of resistance 200 Ω is used to measure voltage across the two resistance separately. Calculate the voltage indicated when voltmeter connected across 100 Ω :-

(1) 1 V

(2) 2 V

(3) 3 V

(4) 4 V

15) A capacitor stores 60μC charge when connected across a battery. When the gap between the plates is filled with a dielectric, a charge of 120μC flows through the battery. The dielectric constant of the material inserted is :

(1) 1

(2) 2

(3) 3

(4) None

16) The time period of a thin bar magnet in uniform magnetic field is T. If the magnet is cut into two equal parts perpendicular to its length, the time period of each part in the same field will be :-

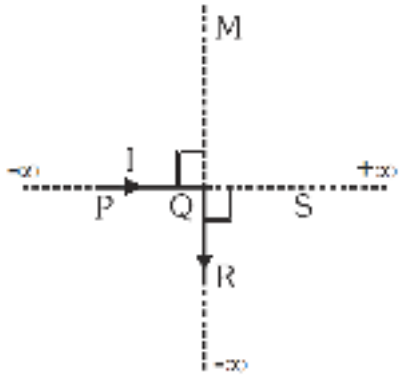
(1) T/2

(2) T

(3) $\sqrt{2}\,T$

(4) 2T

17) An infinitely long conductor PQR is bent to form a right angle as shown. A current I flows through PQR. The magnetic field due to this current at the point M is H_1. Now, another infinitely long straight conductor QS is connected at Q so that the current is $\frac{I}{2}$ in QR as well as in QS, the current in PQ remaining unchanged. The magnetic field at M is now H_2. The ratio $\frac{H_1}{H_2}$ is given by-



(1) $\frac{1}{2}$

(2) 1

(3) $\frac{2}{3}$

(4) 2

18) **Assertion :** Electromagnetic waves carry energy and momentum.
Reason : Electromagnetic waves can be polarised.

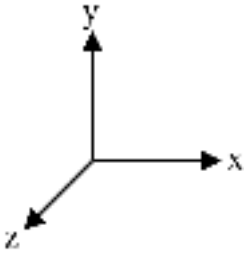
(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true but reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Both assertion and reason are false.

19) Light wave is travelling along y-direction. If the corresponding $\vec{E}$ vector at any time is along the x-axis, the direction of $\vec{B}$ vector at that time is along



(1) y-axis

(2) x-axis

(3) +z-axis

(4) –z-axis

20) The magnification of an object is m_1 when placed 25cm away from a mirror & m_2 when object is

moved 15cm further away from its initial position. If $\frac{m_1}{m_2} = 4$, the focal length of the mirror is :

(1) 10cm

(2) 30cm

(3) 15cm

(4) 20cm

21) Consider following statements for refraction of light through isosceles prism, when angle of deviation is minimum.
(A) The refracted ray inside prism becomes parallel to the base.
(B) Larger angle prisms provide smaller angle of minimum deviation.
(C) Angle of incidence and angle of emergence becomes equal.
(D) There are always two sets of angle of incidence for which deviation will be same except at minimum deviation setting.
(E) Angle of refraction becomes double of prism angle.
Choose the correct answer from the options given below.

(1) A, C and D Only

(2) B, C and D Only

(3) A, B and E Only

(4) B, D and E Only

22) Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A : For a simple microscope, the angular size of the object equals the angular size of the image.
Reason R : Magnification is achieved as the small object can be kept much closer to the eye than 25 cm and hence it subtends a large angle.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) A is true but R is false

(2) Both A and R are true but R is NOT the correct explanation of A.

(3) Both A and R are true and R is the correct explanation of A

(4) A is false but R is true

23) In a certain double slit experimental arrangement interference fringes of width 1.0 mm each are observed when light of wavelength 5000 Å is used. Keeping the set up unaltered, if the source is replaced by another source of wavelength 6000 Å, the fringe width will be :-

(1) 0.5 mm

(2) 1.0 mm

(3) 1.2 mm

(4) 1.5 mm

24) In a single slit diffraction pattern, angular width of central bright maximum, increases when
I. The slit width is decreased

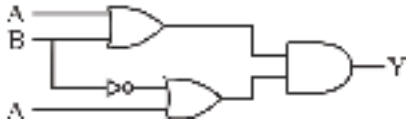
II. The distance between the slit and the screen is increased
III. Light of longer wavelength is used
Select the correct alternative :

(1) I, II, III

(2) I, II only

(3) I, III only

(4) II, III only

25) 

Find out value of Y :-

(1) A

(2) B

(3) A + B

(4) A . B

26) An intrinsic semiconductor has a resistivity of 0.5 Ωm at room temperature. Find the intrinsic carrier concentration if the mobilities of electrons and holes are 0.39 $m^2V^{-1}s^{-1}$ and 0.11 $m^2V^{-1}s^{-1}$ respectively :-

(1) $1.2 \times 10^{18}\ m^{-3}$

(2) $2.5 \times 10^{19}\ m^{-3}$

(3) $1.9 \times 10^{20}\ m^{-3}$

(4) $3.1 \times 10^{21}\ m^{-3}$

27) If momentum (p), area (A) and time (t) are taken to be fundamental quantities, then energy has the dimensional formula :-

(1) $[p^1 A^{-1} t^{-1}]$

(2) $[p^2 A^1 t^1]$

(3) $[p^1 A^{-1/2} t^1]$

(4) $[p^1 A^{1/2} t^{-1}]$

28) A physical quantity σ is given by $\sigma = \frac{3P^2Q^3}{RS^{1/3}}$. The percentage error in the measurements of P, Q, R and S are 2%, 1%, 3%, 6% respectively. Then find out the % error in σ.

(1) 14%

(2) 12%

(3) 16%

(4) 8%

29) If the coefficient of friction between an insect and bowl surface is μ and the radius of the bowl is

r the maximum height to which the insect can crawl in the bowl is :-

(1) $r\left[1-\frac{1}{\sqrt{1-\mu^2}}\right]$

(2) $r\left[1-\frac{1}{\sqrt{1+\mu^2}}\right]$

(3) $\left[1-\frac{r}{\sqrt{1+\mu^2}}\right]$

(4) None of these

30) Match column-I with column-II :-

Column-I		Column-II	
(A)	Definition of force	(p)	Newton's third law
(B)	Measure of force	(q)	Impulse
(C)	Effect of force	(r)	Newton's second law
(D)	Recoiling of gun	(s)	Newton's first law

(1) A-q, B-p, C-r, D-s

(2) A-p, B-q, C-r, D-s

(3) A-s, B-r, C-q, D-p

(4) A-s, B-r, C-p, D-q

31) If the kinetic energy of a body is directly proportional to time t, the magnitude of force acting on the body is :-
(i) directly proportional to $\sqrt{t}$
(ii) inversely proportional to $\sqrt{t}$
(iii) directly proportional to the speed of the body
(iv) inversely proportional to the speed of the body

(1) (i), (ii)

(2) (i), (iii)

(3) (ii), (iv)

(4) (i), (iv)

32) Three particles of masses 10g, 20g and 40g are moving with velocities $10\hat{i}$, $10\hat{j}$ and $10\hat{k}$ (m/s) respectively. If due to some mutual interaction, the first particle comes to rest and the velocity of second particle becomes $\left(3\hat{i}+4\hat{j}\right)$ m/s, then the velocity of third particle is :-

(1) $\hat{i}+\hat{j}+5\hat{k}$

(2) $\hat{j}+10\hat{k}$

(3) $\hat{i}+\hat{j}+10\hat{k}$

(4) $\hat{i}+3\hat{j}+10\hat{k}$

33) A particle of mass m has been thrown up with initial speed u making angle θ with the horizontal.

Find the torque of its weight about the point of projection when it just reached the highest point.

(1) mu $\sin\theta\cdot\cos\theta$

(2) $mu^2 \sin\theta \cos\theta$

(3) $\frac{mu \sin^2\theta}{2}$

(4) $\frac{mu^2 \sin^2\theta}{2}$

34) A solid cylinder is rolling down an inclined plane of height h and reaches the bottom with a linear velocity v. If the mass of the cylinder is M and the radius is R, find the radius of gyration (K) about the central axis parallel to the length of the cylinder.

(1) $\frac{v}{R}\sqrt{v^2-2gh}$

(2) $\frac{R}{v}\sqrt{v^2-2gh}$

(3) $\frac{v}{R}\sqrt{2gh-v^2}$

(4) $\frac{R}{v}\sqrt{2gh-v^2}$

35) Two bodies of masses M_1 and M_2 are kept separated by a distance d. The potential at the point where the gravitational field produced by them is zero, is :-

(1) $-\frac{G}{d}\left(M_1+M_2+2\sqrt{M_1M_2}\right)$

(2) $-\frac{G}{d}\left(M_1M_2+2\sqrt{M_1+M_2}\right)$

(3) $-\frac{G}{d}\left(M_1-M_2+2\sqrt{M_1M_2}\right)$

(4) $-\frac{G}{d}\left(M_1M_2-2\sqrt{M_1+M_2}\right)$

36) Eight drops of a liquid of density ρ and each of radius a are falling through air with a constant velocity 3.75 cms^{-1}. When the eight drops coalesce to form a single drop then terminal velocity of the new drop will be :-

(1) 15×10^{-2} ms^{-1}

(2) 2.4×10^{-2} ms^{-1}

(3) 0.75×10^{-2} ms^{-1}

(4) 25×10^{-2} ms^{-1}

37) Three liquids of densities ρ_1, ρ_2 and ρ_3 (with $\rho_1 > \rho_2 > \rho_3$), having the same value of surface tension T, rise to the same height in three identical capillaries. The angles of contact θ_1, θ_2 and θ_3 obey:-

(1) $\frac{\pi}{2} < \theta_1 < \theta_2 < \theta_3 < \pi$

(2) $\pi > \theta_1 > \theta_2 > \theta_3 > \frac{\pi}{2}$

(3) $\frac{\pi}{2} > \theta_1 > \theta_2 > \theta_3 \geqslant 0$

(4) $0 \leqslant \theta_1 < \theta_2 < \theta_3 < \frac{\pi}{2}$

38) A particle of mass m in a unidirectional potential field have potential energy $U(x) = a + 2bx^2$; where a and b are positive constants. Find time period of oscillation.

(1) $2\pi\sqrt{\frac{2b}{m}}$

(2) $2\pi\sqrt{\frac{m}{2b}}$

(3) $\pi\sqrt{\frac{m}{b}}$

(4) $\pi\sqrt{\frac{b}{m}}$

39) Select the correct statement(s) :-
(A) Number of nodes equals to number of antinodes in closed organ pipe.
(B) In open organ pipe, if number of antinodes is m, then number of nodes will be m-1.
(C) If frequency of 4th harmonic of open organ pipe is 400 Hz, then frequency of 2nd overtone of closed organ pipe of same length is 250 Hz.
(D) Time interval between successive maxima or minima (for superposition of two waves) is $\Delta t = \frac{1}{f_1 - f_2}\,\text{sec}.$

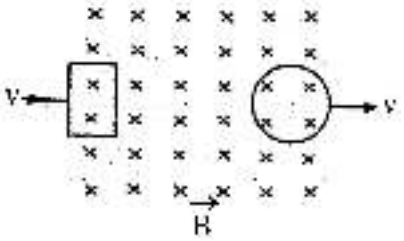
(1) Only (A)

(2) Both (B) and (C)

(3) Only (C)

(4) All of these

40) A rectangular loop and a circular loop are moving out of a uniform magnetic field with a constant velocity v as shown in figure. Then choose correct statements-



(a) Induced emf in rectangular loop remains constant
(b) Induced emf in circular loop remains constant
(c) Induced emf in rectangular loop varies
(d) Induced emf in circular loop varies

(1) a and b

(2) a and d

(3) b and c

(4) c and d

41) A group of electric lamps having a total power consumption of 1000 watt is supplied by an AC voltage, V = 200 sin (310 t + π/3)
If current is I = I_0 sin 310t
Then the rms value of the electric current is :-

(1) 10 amp

(2) $10\sqrt{2}$ amp

(3) 20 amp

(4) $20\sqrt{2}$ amp

42) Photoelectric effect takes place in element A. Its work function is 2.5 eV and required wavelength is λ = 5000 Å. An other element B is having work function of 3eV. Then find out the wavelength that can produce photoelectric effect in B :-

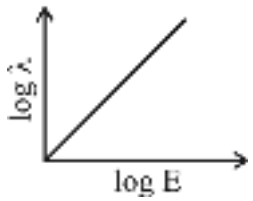
(1) 4100 Å

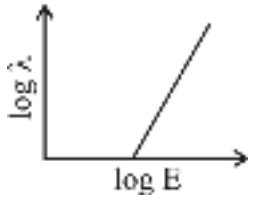
(2) 5000 Å

(3) 6000 Å

(4) 6500 Å

43) The log E - logλ graph between the energy E of an electron and its de-Broglie wavelength λ will be :-

(1) 

(2) 

(3) 

(4) 

44) The acceleration of an electron in the first orbit of the hydrogen atom (n = 1) is :-
(Here symbols have their usual meaning)

(1) $\frac{h^2}{4\pi m^2 r^3}$

(2) $\frac{h^2}{8\pi^2 m^2 r^3}$

(3) $\frac{h^2}{\pi^2 m^2 r^3}$

(4) $\frac{h^2}{4\pi^2 m^2 r^3}$

45) A nucleus X, initially at rest undergoes alpha decay according to the equation. :-

$^{A}_{92}X \rightarrow ^{228}_{Z}Y + \alpha$ + Q

Given that : m(Y) = 228.03 u; $m\left(^{1}_{0}n\right)$ = 1.009 u

$m\left(^{4}_{2}He\right)$ = 4.003 u ; $m\left(^{1}_{1}H\right)$ =1.008 u and

c^2 = 931.5 MeV/u, Q = 5.344 MeV

Binding energy (in MeV) for parent nucleus X is :-

(1) 1728.4

(2) 1823.2

(3) 1900.5

(4) 1680.8

CHEMISTRY

1) Calculate the bond length of C – X bond if C–C bond length is 1.54 Å and X – X bond length is 1.2 Å and EN of C and X are 2 and 3 respectively

(1) 2.74 Å

(2) 1.37 Å

(3) 1.46Å

(4) 1.28Å

2) Match the following electronic configurations with their properties :

(a)	$2s^2 2p^6$	(i)	Smallest atomic radius
(b)	$3s^2 3p^5$	(ii)	Highest IP
(c)	$3s^1$	(iii)	Highest EA
(d)	$2s^2 2p^5$	(iv)	Most metallic

(1) a-(i), b-(iv), c-(ii), d-(iii)

(2) a-(ii), b-(iii), c-(iv), d-(i)

(3) a-(i), b-(iv), c-(iii), d-(ii)

(4) a-(iv), b-(ii), c-(iii), d-(i)

3)

The incorrect order of bond angle :-

(1) $CO_2 > CO_3^{2-} > CF_2Cl_2$

(2) $BF_3 = BCl_3 = BBr_3$

(3) $NO_2^+ > NO_3^- > NO_2$

(4) $PH_3 > AsH_3 > SbH_3$

4) Correct order of bond length is :

(1) $CO_3^{-2} > CO_2 > CO$; C–O bond length

(2) $O_2 < O_2^- < O_2^{-2}$; O–O bond length

(3) $O_2 < O_3 < H_2O_2$; O–O bond length

(4) All of the above

5) **Assertion** :- The dipole moment of CH_3F is lower than that of CH_3Cl.
Reason :- C – F bond is more polar than C – Cl bond.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

6) Match the following

	Pair of species		**Similar properties**
(A)	SiO_2 & $SOCl_2$	(P)	Hybridisation
(B)	$XeOF_2$ & ClF_3	(Q)	Lone pair (at central atom)
(C)	SO_2 & SO_3	(R)	No. of $p_\pi - p_\pi$ bonds
(D)	$XeOF_4$ & SOF_2	(S)	Planar
		(T)	Polar

(1) A→P,R; B→P,S,T; C→P,R,S; D→T

(2) A→P,R; B→P,S,T; C→P,R,S; D→S,T

(3) A→P; B→P,Q,S,T; C→P,R,S; D→Q,T

(4) A→P,Q; B→P,S; C→R,S; D→S,T

7)

Which acid can reduce $AgNO_3$ into metallic silver :-

(1) Hypophosphorous acid

(2) Orthophosphoric acid

(3) Pyrophosphoric acid

(4) Hypophosphoric acid

8) Match the column :

	Metal		Electronic configuration
(A)	Nd	(i)	$4f^{14}\ 5d^{1}\ 6s^{2}$
(B)	Pm	(ii)	$4f^{14}\ 6s^{2}\ 5d^{0}$
(C)	Yb	(iii)	$4f^{4}\ 6s^{2}$
(D)	Lu	(iv)	$4f^{5}\ 6s^{2}$

(1) A-iv; B-iii; C-i; D-iii

(2) A-iii; B-iv; C-i; D-ii

(3) A-iv; B-iii; C-ii; D-i

(4) A-iii; B-iv; C-ii; D-i

9) The wavelength of light absorbed is highest in

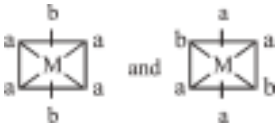
(1) $[Fe(CN)_6]^{3-}$

(2) $[Fe(NH_3)_6]^{3+}$

(3) $[Fe(NO_2)_6]^{3-}$

(4) $[FeCl_6]^{3-}$

10) Which of following statement is incorrect :-

(1) Both geometrical isomers of $[Co(NH_3)_4Cl_2]^{+}$ are optically inactive, whereas  are identical

(2) $[Ca(Edta)]^{-2}$ is used to remove poisonous effect of Pb.

(3) NH_4^{+} (aq) + 2KI + $HgI_2 \rightarrow$ Salt of Millon's base of brown colour.

(4) $CH_2{=}CH_2$ ligand exist in same plane of remaining part of Zeise's salt

11) Which of the following is not correct ?

(1) CFSE in weak ligand field for d^4 configuration is $-0.6\ \Delta_0$

(2) Effective atomic number of Pt in $[PtCl_6]^{2-}$ is 84 (Z of Pt = 78)

(3) Br^{-} is a weak field ligand than H_2O

(4) $[Cr(NH_3)_5NO_2]Cl_2$ shows linkage isomerism

12) Which is incorrect statement ?

(1) Hg_2Cl_2 does not give chromyl chloride test

(2) H_2S not give purple colour complex with sodium nitro prusside

(3) $CuSO_4$ give deep blue soluble complex with excess NH_4OH solution

(4) NH_3 gas give blue colour ppt with Nesseler reagent

13)

Choose the correct matching :-

	Column-I (Given Mixture)		Column-II (Colour Precipitate)
(P)	Pb^{+2} + dil HCl + H_2S→	(A)	Rosy Red ppt.
(Q)	Fe^{+3} + NaOH →	(B)	White ppt.
(R)	Ni^{+2} + DMG solution →	(C)	Black ppt.
(S)	Zn^{+2} + KCN solution →	(D)	Brown ppt.

(1) P-C, Q-D, R-A, S-B

(2) P-C, Q-D, R-B, S-A

(3) P-B, Q-C, R-A, S-D

(4) P-D, Q-C, R-A, S-B

14) Which of the statements is not true?

(1) $K_2Cr_2O_7$ solution in acidic medium is orange

(2) $K_2Cr_2O_7$ solution becomes yellow on increasing the pH beyond 7

(3) On passing H_2S through acidified $K_2Cr_2O_7$ solution, a milky colour is observed

(4) The bond angle of Cr-O-Cr < O-Cr-O in $K_2Cr_2O_7$

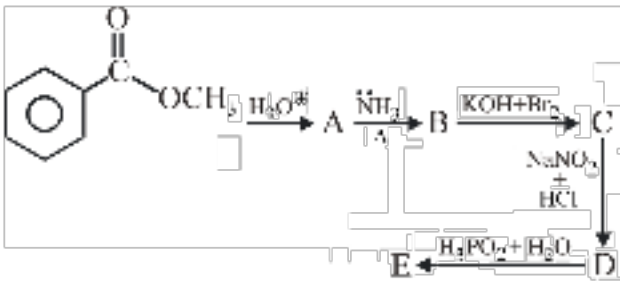
15) Which of the following order is not affected by Lanthanoid contraction ?
(a) Mn $\simeq$ Zn (Size)
(b) Hf > Ti (Ionisation energy)
(c) $Zr^{+4} \simeq Hf^{+4}$ (Ionic size)
(d) Y < La (Size)

(1) a & b

(2) a & c

(3) b & c

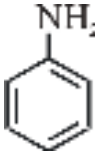
(4) a & d

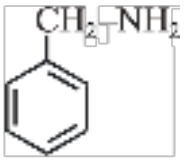


16)
Product (E) is :-

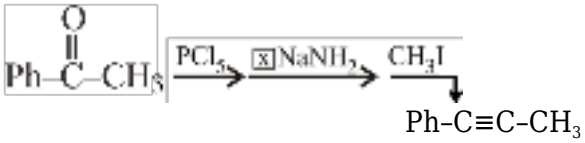
(1) 

(2) 

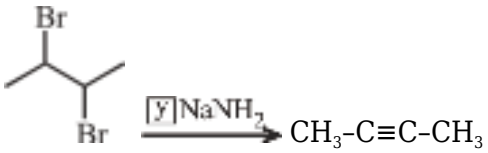
(3) 

(4) 

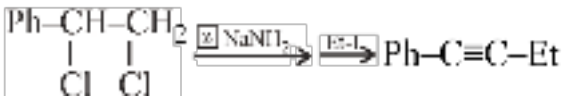
17) Reaction-I



Reaction-II



Reaction-III



x, y & z are moles of used $NaNH_2$. What is x + y + z ?

(1) 6

(2) 8

(3) 5

(4) 7

18) In following conversion

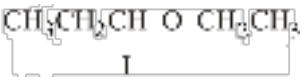
$CH_3CH{=}CH{-}O{-}CH_2CH_3 \xrightarrow[\text{heat}]{\text{Conc. HI}}$

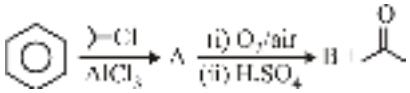
major product is :

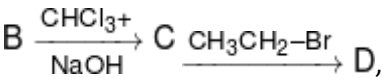
(1) $CH_3CH{=}CHI$ and CH_3CH_2I

(2) $CH_3CH =$ CHI and CH_3CH_2OH

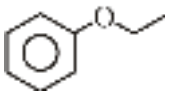
(3) CH_3CH_2CHO and CH_3CH_2I

(4) 

19) 

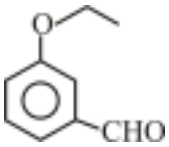


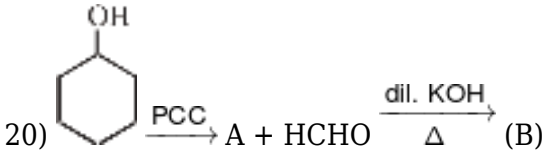
Final Product "D" is :-

(1) 

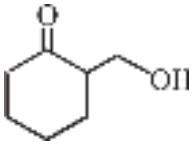
(2) 

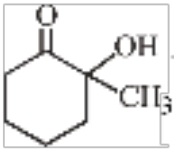
(3) 

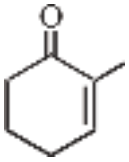
(4) 

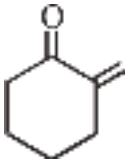
20) 

B is :

(1) 

(2) 

(3) 

(4) 

21) How many structural isomers are possible for the substituted Benzene derivatives with molecular formula C_9H_{12} is –

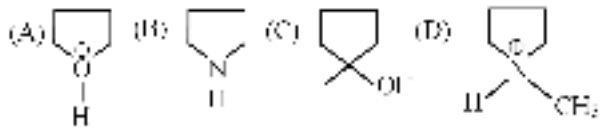
(1) 7

(2) 8

(3) 9

(4) 6

22) Rank the following in decreasing order of acidic strength?

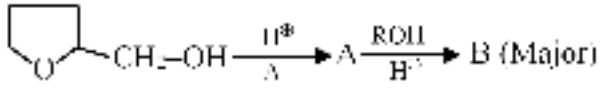


(1) A > C > B > D

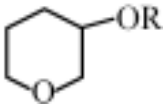
(2) A > D > C > B

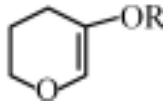
(3) A > D > B > C

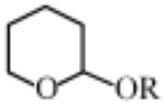
(4) D > A > C > B

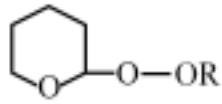
23) 

Major product (B) is –

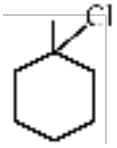
(1) 

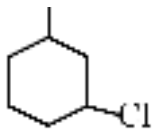
(2) 

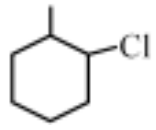
(3) 

(4) 

24) Which of the following alkyl halide gives only one product [excluding stereoisomer] when undergo E_2 reaction –

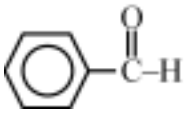
(1) 

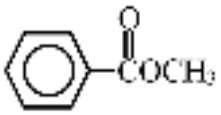
(2) 

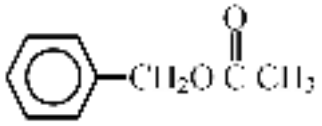
(3) 

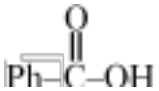
(4) 

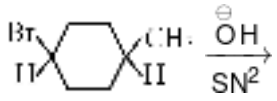
25) Which of the following compound does not form alcohol while treating with grignard reagent

(1) 

(2) 

(3) 

(4) 

26)  Product, Product is –

(1) 

(2) 

(3) 

(4) Both 1 and 3

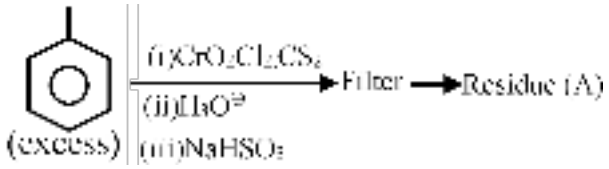
27) Mixture of 1g each of chlorobenzene, aniline and benzoic acid is dissolved in 50 ml ethyl acetate and placed in a separating funnel, 5 M NaOH (30 m) was added in the same funnel. The funnel was shaken vigorously and kept aside. The ethyl acetate layer in the funnel contains -

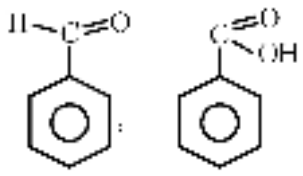
(1) Benzoic acid

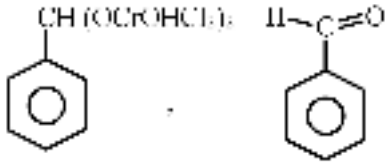
(2) Benzoic acid and aniline

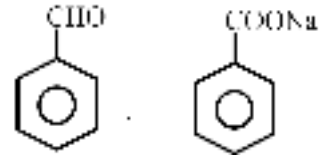
(3) Benzoic acid and chloro benzene

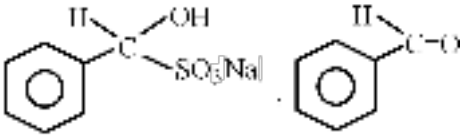
(4) Chloro benzene and aniline

28)  Residue (A) + HCl (dil.) → compound (B)
Structure of residue (A) and compound (B) formed respectively is –

(1) 

(2) 

(3) 

(4) 

29) Match **list-I** with **list-II**.

	List I (Purification technique)	**List II (Mixture of organic compounds)**	
A	Distillation (Simple)	I	Diesel + Petrol
B	Fractional distillation	II	Aniline + Water
C	Steam Distillation	III	Glycerol + Spent lye
D	Distillation under reduced pressure	IV	Chloroform + Aniline

(1) A-IV, B-II, C-I, D-III

(2) A-II, B-III, C-IV, D-I

(3) A-IV, B-I, C-II, D-III

(4) A-II, B-I, C-IV, D-III

30) The α-halix and β-pleated sheet structures of proteins are associated with its

(1) Quarternary structure

(2) Primary structure

(3) Secondary structure

(4) Tertiary structure

31) Two elements X and Y combine to form two compound X_2Y_3 and XY_2. If 3.01×10^{23} molecules of X_2Y_3 weigh 85 gm and atomic mass of Y is 30, then find number of moles in 5 gm of XY_2

(1) 0.02

(2) 0.05

(3) 0.1

(4) 0.5

32) The molality of a solution prepared by dissolving 40 mL of pure methanol (d = 0.8 g mL^{-1}) in 500 g of ethanol is-

(1) 10.8 m

(2) 11.2 m

(3) 2 m

(4) 10.4 m

33) Lattice energy and hydration energy for an ionic solid AB are +210 kJ mol^{-1} and –287 kJ mol^{-1} respectively. At which of the following temperature will AB show maximum solubility :-

(1) 50°C

(2) 60°C

(3) 40°C

(4) 70°C

34) How many coulomb are required for the oxidation of 1 mol of H_2O_2 to O_2?

(1) 9.65×10^4 C

(2) 93000 C

(3) 1.93×10^5 C

(4) 19.3×10^2 C

35) The ratio of the difference in energy between the first and the second Bohr orbit and difference in energy between the second and the third Bohr orbit is-

(1) $\frac{1}{2}$

(2) $\frac{1}{3}$

(3) $\frac{4}{9}$

(4) $\frac{27}{5}$

36) Gadolinium-153, which is used to detect osteoporosis (porous bones), has a half life of 243 days. Which value is closest to the percentage of Gd-153 left in a patient's system after 2 years if its decay follows first order ?

(1) 33%

(2) 26%

(3) 12.5%

(4) 6.25%

37) How many moles of hydrazine (N_2H_4) can be oxidized into N_2 using 2/3 mole of bromate ions $(BrO_3^- \rightarrow Br^-)$:-

(1) $\frac{1}{3}$

(2) 1

(3) 1.5

(4) $\frac{2}{3}$

38)

$Zn + 2H^+ \rightarrow Zn^{2+} + H_2$
If rate law of reaction is $r = k[Zn][H^+]^2$ and pH is decreased from 3 to 2, then rate becomes :-

(1) $\frac{3}{2}$ times

(2) $\frac{1}{100}$ times

(3) 100 times

(4) Unchanged

39) The equilibrium, $NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$, is followed to set-up at 127°C in a closed vessel. The total pressure at equilibrium was 20 atm. The K_c for the reaction is-

(1) 0.092 M^2

(2) 1.085 M^2

(3) 3.045 M^2

(4) 4.14 M^2

40) **Assertion (A) :** Raoult's law can be considered as a special case of Henry's law.
Reason (R) : Both ideal and non-ideal solutions obey Raoult's law.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Assertion is incorrect but Reason is correct.

41) How much current of 60% current efficiency is required to deposit all copper ions from 250 ml of 1 M $CuSO_4$ solution in 193 minute.

(1) 6.944 amp

(2) 69.44 amp

(3) 3.472 amp

(4) 34.72 amp

42) Solubility of AgI in presence of 0.05 M BaI_2 is 10^{-15} M. Solubility of AgI is pure water will be :-

(1) 25×10^{-7} M

(2) 10^{-7} M

(3) 5×10^{-8} M

(4) 10^{-8} M

43) At 700 K if $\Delta G = -11.5$ kJ, what is the value of $\Delta S_{(universe)}$

(1) $-11.5\ kJK^{-1}$
(2) $-16.4\ J\ K^{-1}$
(3) $16.4\ J\ K^{-1}$
(4) $-145\ J\ K^{-1}$

44) **Statement-I :** Starch sol is an example of lyophillic sol.
Statement-II : Conversion of freshly prepared precipitate into a colloidal sol using suitable electrolyte is called coagulation.

(1) Both Statement-I and Statement-II are correct.
(2) Both Statement-I and Statement-II are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

45) What is the pH of a solution prepared by mixing 20 mL of 0.250M sodium formate with 20 mL of 0.250M HCOOH (pK_b for $HCOO^-$ = 10.25) :-

(1) 3.75
(2) 4.18
(3) 5.22
(4) 6.81

BIOLOGY

1) Which type of vascular bundles are found in dicot stem ?

(1) Conjoint Collateral, closed
(2) Radial, open
(3) Conjoint Collateral, open
(4) Radial, Closed

2) Identify the incorrect statement –

(1) All tissues on the inner side of the endodermis such a pericycle, vascular bundles, and pith constitute the stele.
(2) Pith is large and well developed in monocot root.
(3) Monocot roots do not undergo any secondary growth.
(4) The monocot stem posses, parenchymatous hypodermis.

3) Exalbuminous seed store their reserve food mainly in :-

(1) Cotyledon

(2) Seed coat

(3) Endosperm

(4) Perisperm

4) Identify the correct match from the column I, II and III and select the correct option :-

Column-I		Column-II		Column-III	
A	Actinomorphic	i	Bilateral symmetry	a	Mustard
B	Perigynous	ii	Inferior ovary	b	Rose
C	Zygomorphic	iii	Radial symmetry	c	Pea
D	Epigynous	iv	Half-inferior ovary	d	Guava

(1) A-i-a ; B-ii-b ; C-iii-c ; D-iv-d

(2) A-iii-c ; B-iv-d ; C-i-a ; D-ii-b

(3) A-iii-a ; B-iv-b ; C-i-c ; D-ii-d

(4) A-i-c ; B-ii-d ; C-iii-a ; D-iv-b

5) Arrangement of ovules within ovary is known as placentation. The placentation are of different type. Given below are two statements for above structure.
Statement-I : In marginal placentation placenta forms a ridge along ventral suture of ovary and ovules are born on the ridge forming two rows.
Statement-II : Mustard and Argemone have unilocular ovary but it later on become bilocular due to formation of septa of true type during placentation.
Select the correct option in the light of above statement.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

6) Which of the following statements is incorrect about double fertilization ?

(1) Double fertilization is an event unique to flowering plants.

(2) After double fertilization, a mature ovule has 1 diploid & 1 triploid cell.

(3) The central cell after triple fusion become the primary endosperm cell.

(4) Out of two, one male gamete move towards the haploid secondary nucleus & get fused.

7) Which cells of the ovules have abundant reserve food materials?

(1) Hilum

(2) Integument

(3) Nucellus

(4) Funicle

8) Residual persistent Nucellus in some seeds such as blackpepper, beat, and castor is represented by?

(1) Integument
(2) Perisperm
(3) Ovule
(4) Hilum

9) Biological concept of species was given by:

(1) Aristotle
(2) Ernst Mayr
(3) John Ray
(4) Carolus Linnaeus

10) Which of the following statement is incorrect regarding amoeboid protozoans ?

(1) These organism live in fresh water, sea water or most soil.
(2) They move and capture their prey by cilia.
(3) Marine forms have silica shells on their surface.
(4) Some of them are parasite.

11) **Statement I :** Fungi make mycorrhizal association with roots of *Cycas*.
Statement II : Mycorrhizal association facilitate the exchange of nutrients between the fungus and the plants.

(1) Statement-I is correct but Statement-II is incorrect.
(2) Statement-I is incorrect but Statement-II is correct.
(3) Both Statement-I and Statement-II are correct.
(4) Both Statement-I and Statement-II are incorrect.

12) How many member from the following can formes ascospores during sexual reproduction.
(I) *Penicillium* (II) Yeast
(III) *Agaricus* (IV) *Aspergillus*
(V) *Claviceps* (VI) *Neurospora*

(1) 2
(2) 3
(3) 4
(4) 5

13) Select the mismatch :

(1) *Pinus* - Monoecious
(2) *Salvinia* - Heterosporous
(3) *Equisetum* - Homosporous
(4) *Cycas* - Monoecious

14) Match the column-I with column-II

	Column-I (Classes)		Column-II (Example)
I.	Sphenopsida	a.	*Polysiphonia*
II.	Phaeophyceae	b.	*Selaginella*
III.	Lycopsida	c.	*Dictyota*
IV.	Rhodophyceae	d.	*Equisetum*

(1) I-b, II-c, III-d, IV-a

(2) I-b, II-a, III-c, IV-d

(3) I-d, II-c, III-b, IV-a

(4) I-d, II-b, III-c, IV-a

15) **Assertion (A) :** Male gametophytic generation in gymnosperms is highly reduced.
Reason (R) : It is confined to only a limited number of cells.
Options :-

(1) Both (A) and (R) are True and (R) is correct explanation of (A).

(2) Both (A) and (R) are True but (R) is not correct explanation of (A).

(3) (A) is True but (R) is False.

(4) (A) is False but (R) is True.

16) The correct sequence of events in the life cycle of bryophytes is :-
(A) Gametes fused to produce zygote
(B) Formation of sporophyte
(C) Meiotic division to produce haploid spores
(D) Gametophyte produce gametes
(E) The spores Germinate to deveoped into gametophytes.
Choose the correct answer from the options
given below.

(1) D,A,C,B,E

(2) A,B,E,D,C

(3) B,A,C,E,D

(4) D, A, B,C,E

17) Which of the following is most abundant protein on the earth?

(1) Collagen

(2) Trypsin

(3) RuBisCO

(4) Insulin

18)

Match **List-I** with **List-II** with respect to average composition of cells.

List-I		List-II	
A	Nucleic acid	I	1%
B	Proteins	II	3%
C	Ions	III	10-15%
D	Carbohydrates	IV	5-7%

(1) A-IV ; B-III ; C-II ; D-I

(2) A-IV ; B-III ; C-I ; D-II

(3) A-III ; B-IV ; C-I ; D-II

(4) A-IV ; B-II ; C-I ; D-III

19) Which of the following statement is **not correct** ?

(1) The rate of chemical reaction doubles or decreases by half for every 10°C change in either direction.

(2) In the absence of any enzyme, CO_2 and H_2O form 200 molecules of H_2CO_3 in an hour.

(3) Inorganic catalysts work efficiently at high temperatures and high pressures.

(4) Catalytic nature of enzyme is due to its secondary structure only.

20) Which of the following statements is **incorrect**?

(1) Accessory pigments enable a wider range of wavelength of light to be utilised for photosynthesis

(2) Chlorophyll 'a' is visualised as bright/blue green coloured on chromatogram.

(3) The maximum absorption by Chlorophyll 'a' is found in the blue and red regions

(4) Chlorophyll 'b' is the chief pigment associated with photosynthesis.

21) Match **list-I** with **list-II**.

	List-I		List-II
(A)	Accessory pigments	(I)	Yellow green in chromatogram
(B)	Most of the photosynthesis	(II)	Protect chlorophyll 'a' from photo-oxidation
(C)	Paper chromatography	(III)	Blue & Red light
(D)	Chlorophyll-b	(IV)	Separation of pigments

Choose the **correct** answer from the options given below :

(1) A-III, B-I, C-II, D-IV

(2) A-III, B-IV, C-II, D-I

(3) A-II, B-III, C-IV, D-I

(4) A-II, B-III, C-I, D-IV

22) **Statement-I :** The breaking of C—C bonds of complex compounds through oxidation within the cells leading to release of considerable amount of energy is called respiration.
Statement-II : The compounds that are oxidised during the respiration are known as respiratory substrates.

(1) Both Statement-I and Statement-II are correct
(2) Both Statement-I and Statement-II are incorrect
(3) Statement-I is correct and Statement-II is incorrect
(4) Statement-I is incorrect and Statement-II is correct

23) Which of the following statements is **correct?**

(1) All living organisms retain the enzymatic machinery to partially oxidise glucose without the help of oxygen.
(2) The conversion of 3-phosphoglyceric acid (3-PGA) to 2-phosphoglyceric acid (2-PGA), is an ATP yielding process.
(3) The complete oxidation of pyruvate by the stepwise removal of all the hydrogen atoms, leaving three molecules of O_2.
(4) Cytochrome 'c' is a small protein attached to the outer surface of the inner membrane of mitochondria and acts as a mobile carrier for transfer of electrons between complex II and III.

24) Which of the following are **true** about fermentation process?
(A) Pyruvic acid is the final hydrogen acceptor in lactic acid fermentation.
(B) Decarboxylation takes place during lactic acid fermentation.
(C) Equal amount of ATP is synthesised in both lactic acid fermentation and alcohol fermentation.
(D) Both reduction and decarboxylation process take place during alcohol fermentation.

(1) A, B, C and D
(2) B, C and D only
(3) A, B and C only
(4) A, C and D only

25) **Assertion :-** Ethylene helps the plant to increase their absorption surface area.
Reason :- Ethylene promotes root growth and root hair formation.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is correct but Reason is incorrect.
(4) Assertion is incorrect but Reason is correct.

26) Which of the following is a function of auxin ?

(1) Inhibition of cell division
(2) Promotion of cell elongation
(3) Inhibition of flowering in pineapple
(4) Inhibition of apical dominance

27) Given below are two statements :

Statement-I : In monohybrid cross Mendel found that the F_1 always resembled either one of the parents and that the trait of the other parent was not seen in them.
Statement-II : In a typical test cross an organism showing a dominant phenotype is crossed with the recessive parent instead of self-crossing.

(1) Both statement-I and statement-II are false.
(2) Both statement-I and statement-II are true.
(3) Statement-I is true but statement-II is false.
(4) Statement-I is false but statement-II is true.

28) Which of the following is incorrect combination ?

(1) Colourblindness - Sex linked recessive
(2) Sickle cell anaemia - Autosome linked recessive
(3) Phenylketonuria - Autosome linked dominant
(4) Thalassemia - Autosome linked recessive

29) Which of the following statement is not true with respect to Polygenic Inheritance ?

(1) Traits which are generally controlled by three or more genes and are called polygenic traits.
(2) Besides the involvement of multiple genes polygenic inheritance does not take into account the influence of environment.
(3) Human skin colour is another classic example of polygenic inheritance.
(4) In a polygenic trait the phenotype reflects the contribution of each alleles i.e., the effect of each allele is additive.

30) **Statement-I :** Pedigree is a record of some genetic characters or diseases for two or more generation in a family.
Statement-II : Pedigree analysis provides valuable information's regarding genetical make up of human beings.

(1) Both Statement-I and Statement-II are correct.
(2) Both Statement-I and Statement-II are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

31) Stem height of tobacco plant with genotype aaBbCc is 30 cm and the minimum is 20 cm. Find out the height of plant with genotype AABBCC.

(1) 30 cm
(2) 40 cm
(3) 60 cm
(4) 50 cm

32) Which of the following is not related to DNA ?

(1) Phosphodiester bond

(2) N-glycosidic bond

(3) Deoxythymidylic acid

(4) Deoxyuridylic acid

33) Which of the following enzyme facilitate opening of DNA helix at the time of transcription ?

(1) DNA polymerase

(2) DNA helicase

(3) RNA polymerase

(4) RNA ligase

34) In a ds DNA sequence of one strand is 5' ATGAGCTTCCGTCTA 3'. Find out sequence of RNA during transcription ?

(1) 3' UACAGCTTCCGTCTA 5'

(2) 5' UACAGCTTCCGTCTA 3'

(3) 3' AUGAGCUUCCGUCTA 5'

(4) 5' AUGAGCUUCCGUCUA 3'

35) From the following identify the correct combination of salient features of the genetic code.

(1) Universal, Non ambiguous, Overlapping

(2) Degenerate, Over lapping, comma less

(3) Universal, Unambiguous, Degenerate

(4) Degenerate, Nonoverlapping, ambiguous

36) **Statement-I :** Severo Ochoa enzyme was used for enzymatic synthesis of RNA.
Statement-II : Hargobind khorana developed the chemical method for synthesizing RNA molecule.
Select the correct answer from the options given below :

(1) Both statement-I and statement-II are correct.

(2) Only statement-I is correct.

(3) Only statement-II is correct.

(4) Both statement-I and statement-II are incorrect.

37)

Select the correct statements :-
(A) In bacteria, there are three major types of RNAs.
(B) tRNA brings amino acids and reads the genetic code during transcription
(C) All three RNAs (mRNA, tRNA, rRNA) are needed to synthesise a protein in the cell.
(D) rRNA, reads the genetic code during translation.
(E) 'rRNA' play structural and catalytic role during transcription
Choose the most appropriate answer from the options given below :-

(1) A and B only

(2) B and D only

(3) B and E only

(4) A and C only

38) **Statement-I :** Competition occurs when closely related species compete for the same resources that are limited
Statement-II : If two species compete for the same resources they could avoid competition by resource partitioning.

(1) Both **Statement-I** & **II** are true.

(2) **Statement-I** is false and **Statement-II** is true

(3) **Statement-I** is true and **Statement-II** is false

(4) Both **Statement-I** & **II** are false.

39) Given below are two statements :
Statement (I) : When Darwin spoke of the struggle for existence and survival of the fittest in nature. He was convinced that interspecific competition is potent force in organic evolution.
Statement (II) : Resources need not to be limiting for competition to occur, in interference competition, the feeding efficiency of one species might be reduced due to interfering and inhibitory presence of the other species.
In the light of above statement select the correct options :-

(1) Both **Statement I** and **Statement II** are correct.

(2) Both **Statement I** and **Statement II** are incorrect.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) **Statement I** is incorrect but **Statement II** is correct.

40) The algal bloom in a pond ecosystem can be an example of :-

(1) Parasitism

(2) Amensalism

(3) Commensalism

(4) Predation

41) What percentage of photosynthetically active radiation (PAR) is captured by plants?

(1) 50%

(2) 10%

(3) 2-10%

(4) 90%

42) Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): In most ecosystems all the pyramids of number and biomass are upright.
Reason (R): In a food chain generally producers are more in number and biomass than the herbivores and herbivores are more in number and biomass than carnivores.
In the light of the above statements choose the correct answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is NOT the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

43)

How many statements are true.
(a) India has 2.04 % of world area.
(b) India has 8.11% of global species diversity
(c) India is one of the 12 mega diversity countries of the world.
(d) In India 45000 species of animals and 90,000 species of plants have been recorded.

(1) a, b and d are correct

(2) only d is correct

(3) a, d and c are correct

(4) only c is correct

44) An example of In-situ conservation is ?

(1) Hot spot

(2) Zoological parks

(3) Botanical gardens

(4) Wildlife safari parks

45) Which one aspect is not found to present in biodiversity hot spots ?

(1) High degree of threat

(2) Endemism

(3) Species richness

(4) Low interspecific competition

46) Choose the correct option which is common among the given animals.

(a)  (b) 

(c)  (d) 

(1) All have open circulatory system and jointed appendages.

(2) All have metameric segmentation.
(3) All have double solid dorsal nerve cord.
(4) Both (1) and (2)

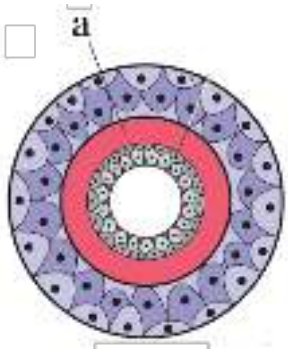
47) True segmentation of body is first observed in the member of which phylum?

(1) Platyhelminthes
(2) Aschelminthes
(3) Annelida
(4) Arthropoda

48) Malphighian tubules are :

(1) Excretory organs of insects
(2) Excretory organs of annelids
(3) Respiratory organs of insects
(4) Respiratory organs of annelids

49) Which is correct about 'a'



(1) Mesoglea
(2) Mesoderm
(3) Ectoderm
(4) Endoderm

50) Find out correct statement for cockroach.

(1) Body wall consist of calcareous shell.
(2) Body cavity is pseudocoelom
(3) Coelom found in cavity of gonads
(4) Body cavity is filled with blood which is red in colour

51) Match the following columns with respect to frog :

	Column-I		**Column-II**
A	Simple eye	i	Cerebellum
B	Hind brain	ii	Possessing only one unit

C	Fertilization	iii	Urea
D	Excretory product	iv	External

Codes :

	A	B	C	D
1	iv	iii	ii	i
2	ii	i	iv	iii
3	i	ii	iii	iv
4	ii	iii	i	iv

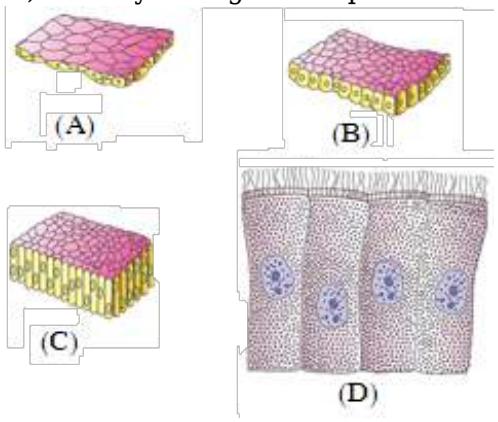
(1) 1

(2) 2

(3) 3

(4) 4

52) Identify the figure of epithelial tissue given below along with it's correct location and function.



		Tissue	Location	Function
1	D	Ciliated epithelium	Bronchioles and fallopian tube	Move particles in specific direction
2	C	Simple Squamous epithelium	Pharynx	Protection
3	B	Keratinised stratified squamous epithelium	Stomach	Secretion & absorption
4	A	Ciliated Columnar epithelium	Intestine	Forming diffusion boundary

(1) 1

(2) 2

(3) 3
(4) 4

53) Thymus gland secrete the peptide hormone called :

(1) Thyroxine
(2) Thymosin
(3) Parathyroid hormone
(4) Adrenaline

54) Androgen act on the _____ and influence the male sexual behaviour.

(1) Peripheral Nervous System
(2) Central Nervous System
(3) Autonomic Nervous System
(4) Visceral nervous system

55)

Glucagon, Insulin, Catecholamines, β-cells, Diabetes mellitus.
How many of the following are related to pancreas?

(1) 3
(2) 2
(3) 4
(4) 5

56) Which part of brain is responsible for urge for eating?

(1) Cerebrum
(2) Hypothalamus
(3) Corpus Callosum
(4) Medulla Oblongata

57) **Assertion :** Outer surface of the axonal membrane possesses a positive charge while its inner surface becomes negatively charged and therefore is polarised.
Reason : Ionic gradients across the resting membrane are maintained by the passive transport of ions by the Na^{+}-K^{+} pump.
Mark the correct choice-

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(2) **Assertion** is true but **Reason** is false.
(3) **Assertion** is false but **Reason** is true.
(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

58) Which type of muscle fibres also called aerobic muscle?

(1) Red muscle fibres

(2) Elastin fibres

(3) Reticular fibres

(4) White muscle fibres

59) Select the correct statements with respect to movement and locomotion:-
Statement (A): Cells of human body exhibit three main types of movements.
Statement (B): Cytoskeletal elements like microfilaments are also involved in amoeboid movement.
Statement (C): Macrophages shows amoeboid movement by forming pseudopodia.

(1) Statement A, B is correct and C is incorrect.

(2) Statement A, C is correct and B is incorrect.

(3) Statement A, B, C all are correct.

(4) Statement B, C is correct and A is incorrect.

60) Match the column I with column II :-

	Column-I		**Column-II**
(A)	Angina	(i)	Deposition of calcium, fat, cholesterol
(B)	Atherosclerosis	(ii)	Inflammation of bronchi and bronchioles.
(C)	Asthma	(iii)	Alveolar walls are damaged.
(D)	Emphysema	(iv)	Not enough oxygen is reaching the heart muscles.

(1) A-iii, B-iv, C-ii, D-i

(2) A-iii, B-iv, C-i, D-ii

(3) A-iv, B-i, C-iii, D-ii

(4) A-iv, B-i, C-ii, D-iii

61) How many of the following statements are incorrect ?
A. Plasma is a viscous fluid having straw colour.
B. Plasma has 16-18% proteineous substance.
C. Leucocyte are colourless due to lack of Hemoglobin.
D. Cortex extends between medullary pyramid to form column of bertini.

(1) Two

(2) Four

(3) Three

(4) One

62) Hypertension is the term for :-

(1) Blood pressure lower than normal

(2) Blood pressure higher than normal

(3) Blood pressure equal to the normal

(4) Stopping of heart beat.

63) **Assertion :** JGA is responsible to maintain GFR.
Reason : Increase in GFR activates the JG cells to release renin.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

64) Which excretory material is most toxic ?

(1) NH_3
(2) Urea
(3) Uric acid
(4) Guanine

65) Every 100 ml of deoxygenated blood delivers approximately _____ ml of CO_2 to Alveoli.

(1) 5 ml
(2) 20 ml
(3) 4 ml
(4) 2 ml

66) Capacitation of sperm is provided by :-

(1) Urethra
(2) Vas deferens
(3) Cervix
(4) Seminal vesicle

67)

Secretions of male accessory glands constitute the seminal plasma which is rich in :

(1) Calcium
(2) Fructose
(3) Enzymes
(4) All of these

68) The embryo with 8 to 16 blastomeres is called :-

(1) Morula
(2) Blastula
(3) Gastrula
(4) Foetus

69) How many of the following are correctly matched?
(i) After one month-Heart formed
(ii) End of second month-Limbs and digits are formed
(iii) First trimester-Major organ systems are formed
(iv) During fifth month-First movement
(v) End of second trimester-Fine hairs on body
Find the correct option :-

(1) 3
(2) 1
(3) 5
(4) 2

70) Identify the correct statement :-

(1) Secondary follicles are surrounded by granulosa cells and theca layer
(2) Primary follicle is characterised by antrum.
(3) Graffian follicles ruptures to release tertiary oocytes.
(4) Primary follicle is surrounded by granulosa cells and theca layer.

71) Which of the following statement is not correct regarding given contraceptive methods and their actions ?

(1) Barrier methods-prevents fertilization
(2) Hormonal IUDs- prevents implantation
(3) Oral pills-prevents ovulation
(4) Vasectomy-prevents spermatogenesis

72) Read the following statements carefully -
(A) Typhoid fever could be confirmed by widal test.
(B) In severe Pneumonia the lips and finger nail may turn grey to bluish in colour.
(C) Common cold is vector borne disease.
(D) Haemozoin is released by Ruptured WBCs of infected person.
(E) During Allergy, IgA types of antibodies are formed.
How many of above statements are correct ?

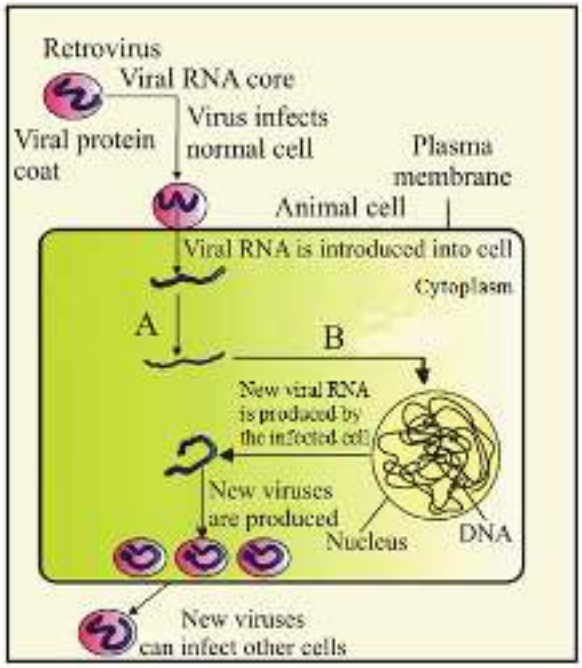
(1) 2
(2) 3
(3) 4
(4) 5

73) **Statement-I :** Vaccines generate memory-B and T cells that recognise the pothogen quickly
Statement-II : Tissue matching and blood group matching are essential before graft/transplant.

(1) Both statement-I and statement-II are correct
(2) Statement-I is correct but statement-II is incorrect
(3) Statement-I is incorrect but statement-II is correct

(4) Both statement-I and statement-II are incorrect

74) Find 'A' & 'B' enzyme



(1) Protease, integrase

(2) Polymerase, integrase

(3) Integrase, polymerase

(4) Reverse transcriptase, Integrase

75) Given below are two statements :
Statement-I : Formation of life was preceded by chemical evolution
Statement-II : First form of life could have come from pre-existing non-living organic molecules.

(1) Only statement-I is correct

(2) Only statement-II is correct

(3) Both statement-I & II are correct

(4) Both statement-I & II are incorrect

76) Choose the incorrect matched pair :-

(1) Homo erectus – 900 cc

(2) Homo habilis – Ate meat

(3) Neanderthal man – Buried their dead

(4) Homo erectus – First man who used fire

77) Read the column-I & II and identify their correct match :

	Column-I		**Column-II**
(A)	Prophase	(i)	Observed clearly under microscope
(B)	Metaphase	(ii)	Formation of mitotic apparatus

(C)	Anaphase	(iii)	Centromeres split
(D)	Telophase	(iv)	Cell plate formation
		(v)	Chiasmata
		(vi)	Homologous chromosome separate
		(vii)	ER reform

(1) A-ii, B-iii, C-vi, D-vii

(2) A-ii, B-i, C-iii, D-vii

(3) A-ii, B-i, C-vi, D-v

(4) A-ii, B-i, C-iii, D-vi

78) Identify the given figure and choose correct answer : 

(1) Homologous chromosome separation

(2) Splitting of the centromere of each chromosome

(3) The bivalent chromosomes align on the equatorial plate

(4) Chromosome get enclosed by a nuclear envelope

79) If mitotically dividing vegetative cell of flowering plant at G_1 stage has 20 chromosome with 20pg DNA then in anaphase cell will have :-

(1) 20 chromosomes/40pg DNA

(2) 40 chromosomes/40pg DNA

(3) 40 chromosomes/80pg DNA

(4) 10 chromosomes/40pg DNA

80) What is false for nucleolus ?

(1) It is not a membrane bound structure

(2) It is a site for active ribosomal RNA synthesis

(3) Clearly visible during M-phase of cell cycle

(4) They become larger and more numerous in actively protein synthesizing cells

81) Identify correct statements from given below-
(I) Several ribosomes may attach to single mRNA and form a chain called polysome.
(II) ETS take place in matrix of mitochondria.
(III) Small, enzyme containing unit membranous structures are called microbodies that present in both plant and animal cell.
(IV) The telocentric chromosome has terminal centromere.
(V) The flagella show 9 + 0 arrangement of microtubules.

(1) I, III & V

(2) II, III & V

(3) I, III & IV

(4) I, IV & V

82)

Match the column.

(a)	Thylakoid	(i)	Site of Krebs cycle reactions
(b)	Mitochondria	(ii)	Composed of cellulose, hemicellulose, pectin and protein
(c)	Cell wall	(iii)	Organised in stacks of grana
(d)	Nucleus	(iv)	Primary site of DNA Replication

(1) c-i, b-iii, d-iv, a-ii

(2) a-iv, b-ii, c-iii, d-i

(3) a-iii, b-i, c-ii, d-iv

(4) a-i, b-iv, c-i, d-ii

83) **Assertion (A) :** Selection of recombinants due to inactivation of antibiotics resistance is a cumbersome procedure.
Reason (R) : Selection due to inactivation of antibiotics requires simultaneous plating on two plates having different antibiotics .
Select the correct option :

(1) Both Assertion and Reason are correct but Reason is NOT the correct explanation of Assertion.

(2) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.

84) **Statement-I :** The recombinant plasmids will lose tetracycline resistance due to insertion of foreign DNA but can still be selected out from non-recombinant ones by plating the transformants on tetracycline containing medium.
Statement-II : A foreign DNA was inserted into the tetracycline resistance gene. Non recombinants would survive on the medium containing ampicillin but not tetracycline.
Choose correct option for given above statements :-

(1) Both **Statement I** and **Statement II** are correct.

(2) Both **Statement I** and **Statement II** are incorrect.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) **Statement I** is incorrect but **Statement II** is correct

85) A piece of DNA if needed to be propagated during recombinant DNA procedures requires a vector, because vector provide :-

(1) Origin of replication site

(2) Origin of transcription site

(3) Origin of translation site

(4) All the above

86) **Assertion (A) :** Ethical standards are required to evaluate the morality of all human activities that might help or harm living organisms.
Reason (R) : Genetic modification of organisms can have predictable results when such organisms are introduced into the ecosystem.

(1) Both Assertion and Reason are correct but Reason is not correct explanation of Assertion.

(2) Assertion is correct but Reason is incorrect.

(3) Assertion is incorrect but Reason is correct.

(4) Both Assertion and Reason are correct and Reason is correct explanation of Assertion.

87) Fill up the blanks. I. Using ....A.... vectors, Nematode specific genes are introduced into the host plant. II. RNAi takes place in all eukaryotic organisms as a method of .....B..... III. The genes .....C.... and .....D.... controls cotton bolloworms.
IV. ....E... has been introduced in Bt corn to protect the same form corn borer.
A to E in above statements can be :-

(1) A-fungal, B-pest resistance,
C-cry II Ab, D-cry II Ab, E-cry I Ac

(2) A-nematode, B-protection against nematodes,
C-cry I Ac, D-cry II Ab, E-cry I Ab

(3) A-*Agrobacterium*, B-cellular defense,
C-cry II Ab, D-cry I Ac, E-cry I Ab

(4) A-virus, B-protection against viral infection,
C-cry I Ac, D-cry II Ab, E-cry II Ac

88) Cultivation of *Bt* cotton has been much in the news. The prefix ‘‘*Bt*’’ means :–

(1) ‘‘Barium – treated’’ cotton seeds

(2) ‘‘Bigger thread’’ variety of cotton with better tensile strength.

(3) Produced by ‘‘biotechnology’’ using restriction enzymes and ligases.

(4) Carrying an endotoxin gene from *Bacillus thuringiensis*.

89) **Statement-I :-** Streptokinase is an enzyme produced by the bacterium *Streptococcus* and modified, is used to dissolve blood clot in treatment of heart attacks.
Statement-II :- Streptokinase act as a clot buster for removing clots from the blood vessels of patients who have undergone heart attack.

(1) Both **Statement I** and **Statement II** are correct.

(2) Both **Statement I** and **Statement II** are incorrect.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) **Statement I** is incorrect but **Statement II** is correct.

90) Lactic acid bacteria (LAB), particularly *Lactobacillus*, are widely used in the preparation of curd and other fermented milk products. During fermentation, these bacteria convert lactose present in

milk into lactic acid. The accumulation of lactic acid lowers the pH, leading to coagulation of milk proteins such as casein. In addition to improving texture and taste, LAB also enhance the nutritional quality of curd by increasing vitamin B_{12} content and inhibiting the growth of harmful microbes in the digestive tract.

Statements:

Statement-1 :- During curd formation, lactic acid bacteria convert lactose into lactic acid, which lowers the pH of milk, causes coagulation of casein proteins, improves digestibility, enhances vitamin B_{12} content, and suppresses the growth of pathogenic microorganisms in the gut.

Statement-2 :- The sour taste and thick consistency of curd are primarily due to the production of carbon dioxide by lactic acid bacteria during aerobic respiration inside milk.

Options:

(1) Both statement-1 and statement-2 are correct

(2) Both statement-1 and statement-2 are incorrect

(3) Statement-1 is correct but statement-2 is incorrect

(4) Statement-1 is incorrect but statement-2 is correct

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	4	4	2	1	2	3	2	4	3	2	3	4	1	3	1	3	2	4	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	2	3	3	1	2	4	2	2	3	3	4	2	4	1	1	4	3	4	2
Q.	41	42	43	44	45															
A.	2	1	3	4	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	2	3	4	2	3	1	4	4	4	2	4	1	4	4	2	2	3	2	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	2	3	4	4	4	4	4	3	3	2	3	3	3	4	3	2	3	1	3
Q.	86	87	88	89	90															
A.	1	4	3	3	1															

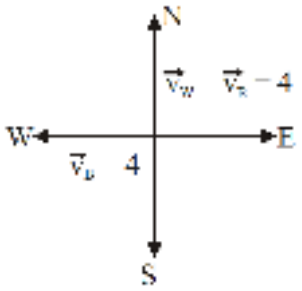
BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	4	1	3	3	4	3	2	2	2	2	4	4	3	1	4	3	2	4	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	1	1	4	1	2	2	3	2	1	4	4	3	4	3	1	4	1	1	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	1	4	1	4	2	3	1	1	3	2	1	2	2	3	2	2	1	3	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	2	3	1	3	3	4	1	3	1	4	1	1	4	3	2	2	3	2	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	3	2	3	1	2	3	4	1	3										

SOLUTIONS

PHYSICS

1) direction of flag hosting is along $-\vec{v}_{BG} + \vec{v}_W$
⇒ which is along North-West



2) At 45°, $v_x = v_y$
⇒ $u_x = u_y - gt$

$$t = \frac{u_y - u_x}{g} = \frac{40(\sin 60^\circ - \sin 30^\circ)}{9.8} = 1.5 \text{ sec.}$$

3) $$\frac{3.91 \times 0.5142}{0.070} = 28.7217$$
from rule of least SF, No of SF in result should be 2
⇒ Result = 29

4) **Solution :**
Bottom :

$$T_b = \frac{mv_b^2}{R} + mg$$

$$v_b^2 = 8gR,\ T_b = m\frac{8gR}{R} + mg = 9\,mg$$

Top :

$$T_t = \frac{mv_t^2}{R} - mg$$

$v_t^2 = 8gR - 4gR = 4gR$, (using conservation of mechanical energy)

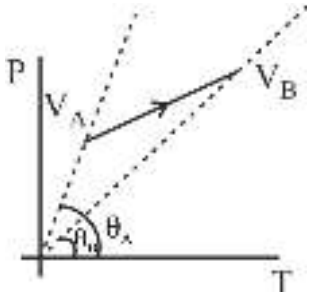
$$T_t = m\frac{4gR}{R} - mg = 4mg - mg = 3mg$$

Thus,
$T_t : T_b = 3mg : 9mg = 1 : 3.$

5) Heat loss by water to 0°C
$Q_1 = 250 \times 1 \times (25 - 0) = 6250$ cal
Heat gain by ice to 0°C
$Q_2 = 2 \times 50 \times 0.5 \times 15 + 2 \times 50 \times 80 = 8750$ cal
$\therefore Q_2 > Q_1$ hence full ice will not melt. Let mass of ice melted is m these
Heat loss = Heat gain
$250 \times 1 \times (25 - 0) = 2 \times 50 \times 0.5\,(0 - (-15)) + m \times 80$

m = 68.75 gm

Hence final mass of water = 250 + 68.75 = $\frac{1275}{4}$gm

6) 

Slope of graph = $\frac{P}{T} = \frac{\mu R}{V}$

Slope $\propto \frac{1}{V}$

$\tan\theta \propto \frac{1}{V}$

Since $\tan\theta_A > \tan\theta_B \Rightarrow V_A < V_B$

7)

$Q = \sigma A\, t\, (T^4 - T_0^4)$

If T, T_0, σ and t are same for both bodies then

$\frac{Q_{sphere}}{Q_{cube}} = \frac{A_{sphere}}{A_{cube}} = \frac{4\pi r^2}{6a^2}$ .....(i)

But according to problem, volume of sphere = Volume of cube $\Rightarrow \frac{4}{3}\pi r^3 = a^3 \Rightarrow a = \left(\frac{4}{3}\pi\right)^{1/3} r$

Substituting the value of a in equation (i) we get

$$\frac{Q_{sphere}}{Q_{cube}} = \frac{4\pi r^2}{6a^2} = \frac{4\pi r^2}{6\left\{\left(\frac{4}{3}\pi\right)^{1/3} r\right\}^2}$$

$$= \frac{4\pi r^2}{6\left(\frac{4}{3}\pi\right)^{2/3} r^2} = \left(\frac{\pi}{6}\right)^{1/3} : 1$$

8)

slope = $\frac{\Delta Q}{\Delta T} = \frac{2500}{100} = 25$

$\Delta Q = du + \Delta w$

$\Delta Q = \mu c_V dT + \mu R dT$

$\frac{\Delta Q}{dT} = \mu c_V + \mu R$

$\mu = 1$

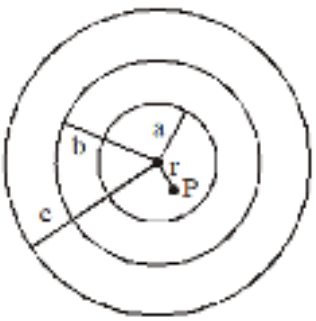
$25 = c_V + R$

$$25 = \frac{R}{\gamma - 1} + R$$
$$25 = R\left(\frac{1}{\gamma - 1} + 1\right)$$
$$25 = \frac{25}{3}\left(\frac{1 + \gamma - 1}{\gamma - 1}\right)$$
$$\frac{\gamma}{\gamma - 1} = 3$$
$\gamma = 3\gamma - 3$
$2\gamma = 3$
$$\boxed{\gamma = \frac{3}{2}}$$

9) $V = \omega\sqrt{A^2 - \frac{A^2}{4}}$
$= \frac{\sqrt{3}}{2}\omega A = \frac{\sqrt{3}}{2}V_{max}$
K.E. $= \frac{3}{4}$ $(K.E.)_{max}$
About P.E., it will be $\frac{1}{4}$ times of its maximum value
$a = -\omega^2\left(\frac{A}{2}\right) = \frac{1}{2}a_{max}$

10)

As drop fall with constant velocity then air resistance = mg = F_A. If uniform electric field is applied vertical to make the oil drop ascend up with the same constant speed.
Fe = mg + F_A = 2mg
⇒ Fe = qE = 2mg
$$\Rightarrow E = \frac{2mg}{q} = \frac{2 \times 1.6 \times 10^{-15} \times 10^{-3} \times 10}{9.6 \times 10^{-19}}$$
$E = \frac{1}{3} \times 10^2$ N/C
If electrostatic force on negative charge is upward then electric field is downwards.



11)
Potential at point P, $V = \frac{kQ_a}{a} + \frac{kQ_b}{b} + \frac{kQ_c}{c}$
$Q_a : Q_b : Q_c : : a^2 : b^2 : c^2$

{since $\sigma_a = \sigma_b = \sigma_c$}

$$\therefore Q_a = \left[\frac{a^2}{a^2+b^2+c^2}\right] Q$$

$$Q_b = \left[\frac{b^2}{a^2+b^2+c^2}\right] Q$$

$$Q_c = \left[\frac{c^2}{a^2+b^2+c^2}\right] Q$$

$$V = \frac{Q}{4\pi\in_0}\left[\frac{(a+b+c)}{a^2+b^2+c^2}\right]$$

 Correct answer is (2)

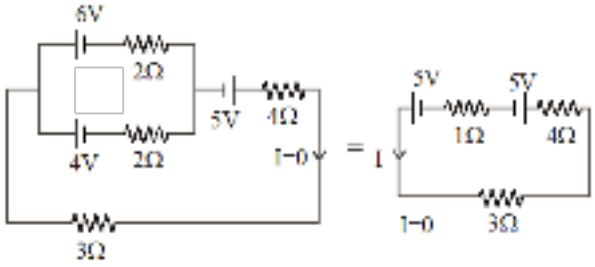
12)

Its a balanced WSB.

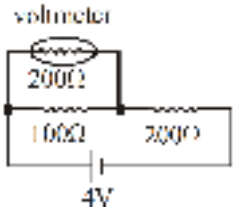
So $\frac{R_3}{R_4} = \frac{R_1}{R_2}$

$R_1R_4 = R_2R_3$

13)



14)



Effective resistance of circuit

$R_{eff} = 200 + \frac{200 \times 100}{200+100} = \frac{800}{3}\Omega$

current in circuit $I_{eff} = \frac{4}{\frac{800}{3}} = \frac{3}{200}$ A

so voltage across the voltmeter = $4 - \left(\frac{3}{200} \times 200\right)$

= 1 volt

Hence option (1)

15)

q_i = 60μc, q_f = 120 μc

 $q_f = Kq_i$

$K = \frac{180}{60} = 3$

16) $T' = \frac{T}{n}$

17) $H_1 = \frac{\mu_0 I}{4\pi d}$

$H_2 = \frac{\mu_0 I}{4\pi d} + \frac{\mu_0 (I/2)}{4\pi d}$

18) Consider a plane perpendicular to the direction of propagation of the electromagnetic wave. If electric charges are present in this plane, they will be set and sustained in motion by the electric and magnetic fields of the electromagnetic wave. The charge thus acquired energy and momentum from the wave. This illustrate the fact that an electromagnetic wave like other waves carries energy and momentum.

19) Direction of wave propagation is given by $\vec{E} \times \vec{B}$.

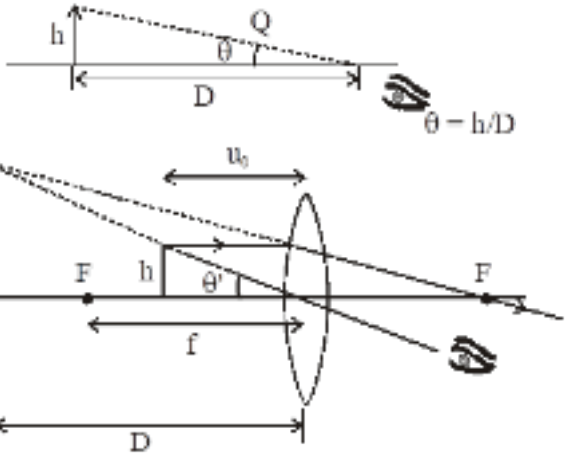
20)

$m = \frac{f}{f-u} \Rightarrow m_1 = \frac{f}{f+25} \quad m_2 = \frac{f}{f+40}$

on dividing $\frac{m_1}{m_2} = \frac{f+40}{f+25} = 4 \Rightarrow f = -20$ cm

21) $\delta = i + e - A$.....(For $\delta_{min} \Rightarrow i = e$)
and refracted ray is parallel to base
A, C, D are correct

22)



$\theta' = \frac{h}{u_0}$; θ' is same for both object and image

$m = \frac{\theta'}{\theta} = \frac{D}{u_0} \Rightarrow u^0 < D \Rightarrow$ Hence m > 1

23) $\beta = \frac{\lambda D}{d} \Rightarrow \frac{\beta_2}{\beta_1} = \frac{\lambda_2}{\lambda_1}$

$\beta_2 = \beta_1 \frac{\lambda_2}{\lambda_1}$

$= 1.0 \times 10^{-3} \times \frac{6000 \times 10^{-10}}{5000 \times 10^{-10}}$

$\beta_2 = 1.2 \times 10^{-3}$m = 1.2 mm

24) $W_\theta = \frac{2\lambda}{a}$

25) $Y = (A + B) \cdot (\bar{B} + A)$

$Y = A\bar{B} + AB + A + B\bar{B}$

$Y = A(1 + B + \bar{B}) = A$

26) $\frac{1}{\rho} = n_i e(\mu_e + \mu_n)$

$n_i = \frac{1}{\rho e(\mu_e + \mu_n)}$

$= \frac{1}{0.5 \times 1.6 \times 10^{-19} \times (0.39 + 0.11)}$

$= 2.5 \times 10^{19}$ m^{-3}

27) Let, energy $E = k\, p^a\, A^b t^c$ ....(i)

where k is a dimensionless constant of proportionality. Equating dimensions on both sides of (i), we get

$[ML^2T^{-2}] = [MLT^{-1}]^a\, [M^0L^2T^0]^b\, [M^0L^0T]^c$

$= [M^aL^{a+2b}T^{-a+c}]$

Applying the principle of homogeneity of dimensions, we get

$a = 1$ ....(ii)

$a + 2b = 2$ ....(iii)

$-a + c = -2$ ....(iv)

On solving eqs. (ii), (iii) and (iv), we get

$a = 1, b = \frac{1}{2}, c = -1$

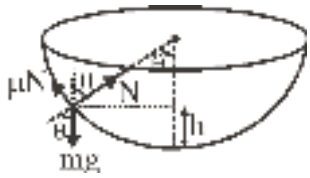
$\therefore [E] = [p^1\, A^{1/2} t^{-1}]$

28) $\sigma = \frac{3P^2Q^3}{RS^{1/3}}$

$\frac{\Delta\sigma}{\sigma} \times 100 = \pm \left[2\frac{\Delta P}{P} + 3\frac{\Delta Q}{Q} + \frac{\Delta R}{R} + \frac{1}{3}\frac{\Delta S}{S}\right] \times 100$

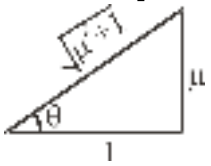
$= \pm$ [4+3+3+2]

= ± 12%

29) 

N = mg cos θ & mg sin θ = μN

⇒ sin θ = μ cos θ ⇒ tan θ =μ



$h = r(1 - \cos\theta) = r\left(1 - \frac{1}{\sqrt{1+\mu^2}}\right)$

30) Newton's first law defines force. Newton's second law gives us a measure of force. Impulse gives us the effect of force. Recoiling of gun is accounted for by Newton's 3rd law.

31) K ∝ t ⇒ K = λt, where λ is a constant

$\frac{1}{2}mv^2 = \lambda t \Rightarrow v = \sqrt{\frac{2\lambda}{m}}t^{\frac{1}{2}}$

$a = \frac{dv}{dt} = \sqrt{\frac{2\lambda}{m}}\frac{1}{2}t^{-1/2}$

$F = ma = \sqrt{\frac{\lambda m}{2}}\frac{1}{t^{1/2}} \Rightarrow F \propto \frac{1}{\sqrt{t}}$, (ii) is O.K.

$F = \sqrt{\frac{\lambda m}{2}}\frac{1}{t^{1/2}} = \sqrt{\frac{\lambda m}{2}} \times \frac{1}{v}\sqrt{\frac{2\lambda}{m}}$

$\Rightarrow F \propto \frac{1}{v}$, (iv) is O.K.

32) From conservation of momentum,

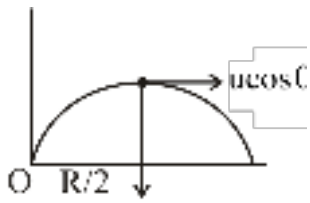
$m_1\vec{u}_1 + m_2\vec{u}_2 + m_3\vec{u}_3 = m_1\vec{v}_1 + m_2\vec{v}_2 + m_3\vec{v}_3$

$10\left(10\hat{i}\right) + 20\left(10\hat{j}\right) + 40\left(10\hat{k}\right)$

$= 10(0) + 20(3\hat{i} + 4\hat{j}) + 40\vec{v}_3$

$\Rightarrow 40\vec{V}_3 = 40\hat{i} + 120\hat{j} + 400\hat{k}$

$\vec{V}_3 = \hat{i} + 3\hat{j} + 10\hat{k}$



33)

$|\vec{\tau}_{\text{about O}}| = (R/2)(mg)$

$= \frac{u^2(2\sin\theta\cos\theta)}{2g} \times mg$

$\Rightarrow mu^2 \sin\theta \cos\theta$

34) $v = \sqrt{\frac{2gh}{1+\frac{K^2}{R^2}}}$

$$v^2 = \frac{2gh}{1+\frac{K^2}{R^2}}$$

$$1+\frac{K^2}{R^2} = \frac{2gh}{v^2}$$

$$\frac{K^2}{R^2} = \frac{2gh}{v^2} - 1$$

$$\frac{K^2}{R^2} = \frac{2gh - v^2}{v^2}$$

$$K^2 = \frac{R^2}{v^2}(2gh - v^2)$$

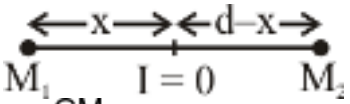
$$K = \frac{R}{v}\sqrt{2gh - v^2}$$

35) $\frac{GM_1}{x^2} = \frac{GM_2}{(d-x)^2} \Rightarrow \frac{d-x}{x} = \sqrt{\frac{M_2}{M_1}}$

$\Rightarrow \frac{d}{x} = 1 + \sqrt{\frac{M_2}{M_1}} \Rightarrow x = \frac{\sqrt{M_1}d}{\sqrt{M_1}+\sqrt{M_2}}$

Now potential,

$-\frac{GM_1}{x} - \frac{GM_2}{d-x}$



$$= \frac{-GM_1}{\sqrt{M_1}d}\left(\sqrt{M_1}+\sqrt{M_2}\right) - \frac{GM_2}{\sqrt{M_2}d}\left(\sqrt{M_1}+\sqrt{M_2}\right)$$

$$= \frac{-G}{d}\left(\sqrt{M_1}+\sqrt{M_2}\right)^2$$

$$= \frac{-G}{d}\left(M_1 + M_2 + 2\sqrt{M_1M_2}\right)$$

36) $R = n^{\frac{1}{3}} r = (8)^{\frac{1}{3}} a = 2a$

$$\frac{v_T'}{v_T} = \left(\frac{R}{r}\right)^2 = \left(\frac{2a}{a}\right)^2 = 4$$

$v'_T = 4\, v_T = 4 \times 3.75$ cm/s $= 15 \times 10^{-2}$ ms^{-1}

37) $h = \frac{2T\cos\theta}{r\rho g}$

here

h, T, r, g = constant

$\square \cos\theta \propto \rho$
$\therefore \rho \uparrow \rightarrow \cos\theta \uparrow \rightarrow \theta \downarrow$

38) $F = -\frac{dU}{dx}$
F = –4bx
$T = 2\pi\sqrt{\frac{m}{k}}$
$T = 2\pi\sqrt{\frac{m}{4b}}$
$T = \pi\sqrt{\frac{m}{b}}$

39)

(A) → Theory based
for COP ⇒ No. of N = No. of AN
(B) For OOP,
No. of N = (No. of AN) – 1
(C) 4th Harmonic of OOP
$\frac{4V}{2\ell} = 400\,Hz \Rightarrow \frac{V}{\ell} = 200$
For COP of same length
2nd OT freq.
$\Rightarrow \frac{5V}{4\ell} = \frac{5}{4} \times 200 = 250\,Hz$
(D) Beat freq. = $f_2 – f_1$ Hz = b
Time interval b/w two successive maxima or two successive minima.
$\Delta t = \frac{1}{b} = \frac{1}{f_2 - f_1}$

40) - **Rectangular loop:** dA/dt is constant → emf remains constant
- **Circular loop:** dA/dt changes with position → emf varies

41)

$P = V_{RMS} I_{RMS} \cos\phi$
$1000 = \frac{200}{\sqrt{2}} (I_{RMS}) \cos 60$
$\Rightarrow I_{RMS} = 10\sqrt{2}\,amp$

42) $\frac{hc}{5000Å} \quad 2.5eV$
⇒ hc = 5000 × 2.5 eV Å
$\phi_B = \frac{hC}{\lambda_0} \Rightarrow \lambda^0 = \frac{5000 - 2.5eVÅ}{3eV}$ = 4133.3 Å

$\Rightarrow \lambda \leq \lambda_0$ can produce photoelectric effect.

43) $\lambda = \frac{h}{\sqrt{2mE}}$

$\lambda = \left(\frac{h}{2m}\right)^{\frac{1}{2}} E^{\frac{-1}{2}}$

$\log \lambda = \log\left(\frac{h}{2m}\right)^{\frac{1}{2}} + \log (E)^{\frac{-1}{2}}$

$\log \lambda = -\frac{1}{2}\log E + \frac{1}{2}\log\left(\frac{h}{2m}\right)$

$y = - mx + C$

44) $mvr = \frac{nh}{2\pi} \Rightarrow v^2r^2 = \frac{n^2h^2}{4\pi^2m^2}$

acceleration

$a = \frac{v^2}{r} = \frac{v^2r^2}{r^3} = \frac{n^2h^2}{4\pi^2m^2r^3}$

put n = 1

45) Q = (BE of y + BE of α) – BE of x

⇒ BE of x = (BE of y) + (BE of α) – Q value

$= (\Delta m_y + \Delta m_\alpha) \times 931.5$ MeV/u – 5.344 MeV

= [(90×1.008 + 138 ×1.009 – 228.03) +

(2 ×1.008 + 2 × 1.009–4.003)] × 931.5 MeV – 5.344 MeV

= (1828.54 – 5.344)MeV = 1823.2 MeV

CHEMISTRY

46)

$\delta = r_A + r_B - 0.09\ \Delta \in N$ equation

47)

Convert configuration in elements like (b) $3s^2 3p^5$ is Cl hence show highest EA.

48) $NO_2^+ > NO_2 > NO_3^-$

180° 134° 120°

49)

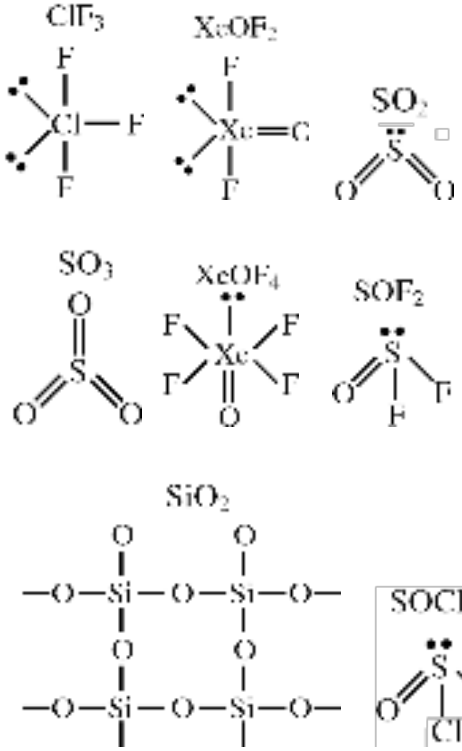
Apply resonance & MOT concept

50)

C-F bond is more polar in CF_3 due to high ΔEN but its dipole moment is less than CH_3Cl.

51)

According to structure



52)

Hypophosphorus acid has 2 P-H bond responsible for reducing nature.

53)

NCERT-XII, Pg. # 110

54)

Weakest the ligand maximum will be absorbed wavelength.

55)

CH_2=CH_2 ligand is perpendicular to remaining part plane.

56)

EAN of Pt is here 78 – 4 + 12 = 86

57)

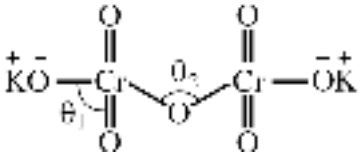
NH_3 with Nesseler reagent gives brown complex.

58)

PbS – Black, $Fe(OH)_3$ – Brown, Ni–dmg–Red, $(Zn(CN)_4)^{-3}$–white

59)

$K_2Cr_2O_7$

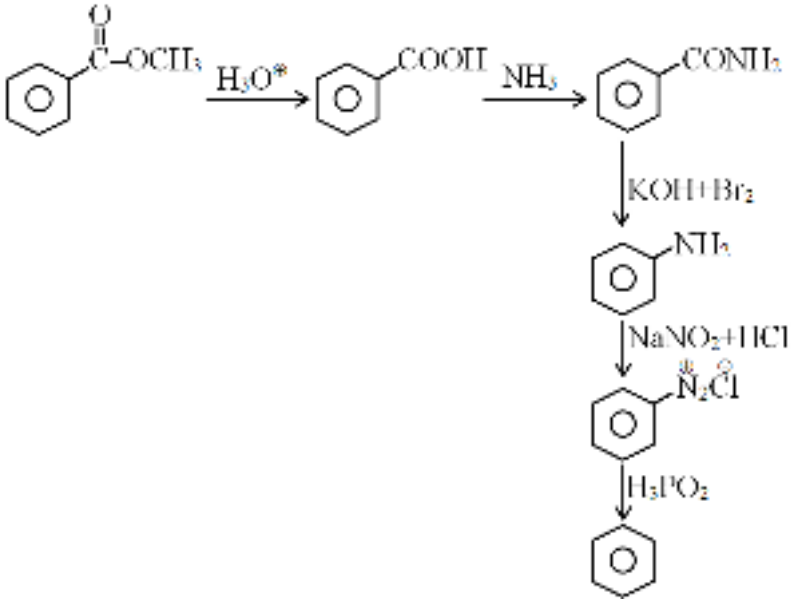


Here $\theta_2 > \theta_1$
126° ≃ 109°28'

60)

Lanthanoids contraction applicable from Hf_{72} onwards.

61)



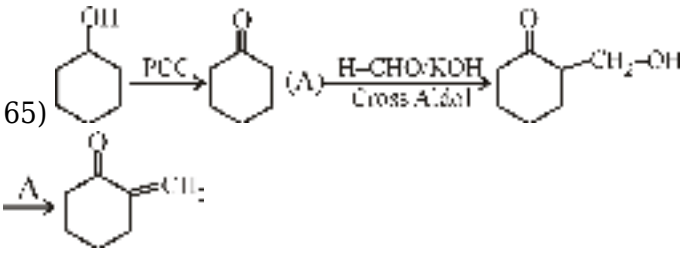
62)

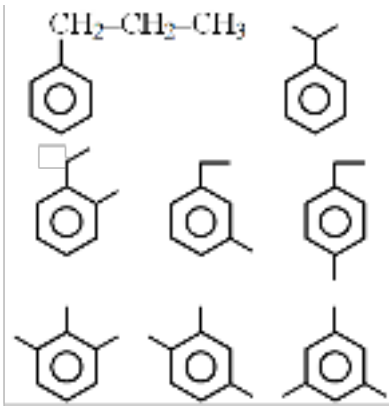
Due to elimination & NSR reaction.

63) CH_3–CH=CH– from unstable carb-cation, so it form alcohol and CH_2–CH_3 group from alkylhalide. We know that OH group is not stable on double bonded carbon it show tautomerism, that why CH_3–CH=CH– group form aldehyde.

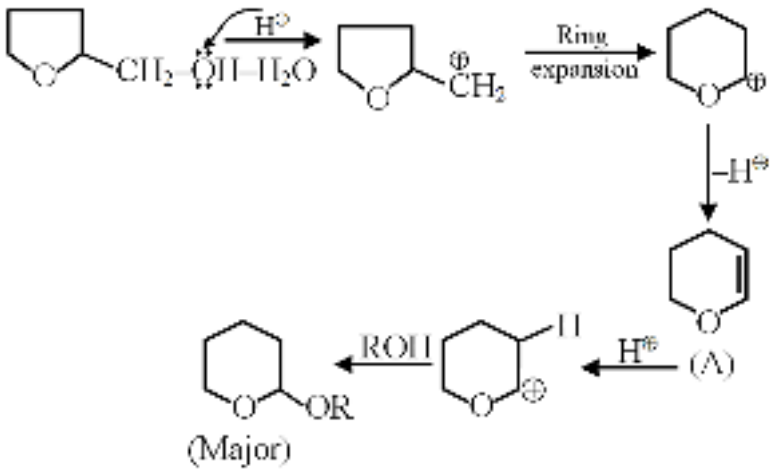
64)

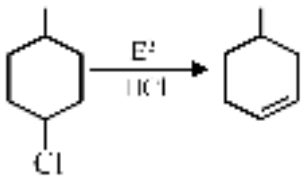
Due Reimer Tiemann reaction followed by NSR

65) 

66) 

67) Due to EN ⇒ $\overset{\oplus}{O} > \overset{\oplus}{N} > O > N$

68) 

69) 

70)

Acid-base neutralization producing a hydrocarbon

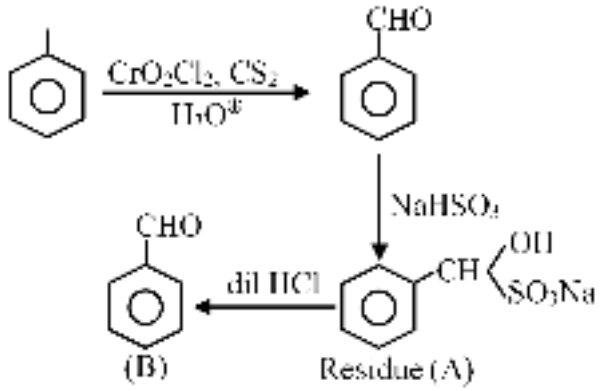
71)

It is an S_N2 reaction.

72)

Chloro benzene and aniline doesn't react with NaOH.

73)



74)

Fact Based

75)

Both are associated with secondary structure.

76) 3.01×10^{23} molecule $X_2Y_3 \rightarrow 85$ gm
1 mole = 6.02×10^{23} molecule
$X_2Y_3 \rightarrow 85 \times 2 = 170$ gm
Also, 1 mole = molar mass of substance
$\therefore$ 2X + 3Y = 170 (X, Y = atomic masses)
X = 40 (since Y = 30)
Molar mass of XY_2 = X + 2Y
= 40 + 2 × (30) = 100

Moles of $XY_2 = \frac{\text{mass}}{\text{molar mass}} = \frac{5}{100} = 0.05$

77) Mass of methanol = d × V = 0.8 × 40 = 32 g

$m = \frac{32/32}{500/1000} = 2$

78) $\Delta H_{dissolution} = (\Delta H)_{lattice} + (\Delta H)_{hydration}$
= (+210) + (–287)

= –ve (Exothermic dissolution)
 S ↓ when T ↑
S = max. when T = min

79) $H_2O_2 \rightarrow O_2 + 2H^+ + 2e^-$
1 mol $H_2O_2 \equiv 2$ mol e^-
$\equiv 2 \times 96500$ C $\equiv 193000$ C
$= 1.93 \times 10^{-5}$ C

80)

$$E_1 - E_2 = -13.6 \times \left(\frac{1}{1^2} - \frac{1}{2^2}\right) = -13.6 \times \frac{3}{4}$$
$$E_2 - E_3 = -13.6 \times \left(\frac{1}{2^2} - \frac{1}{3^2}\right) = -13.6 \times \frac{5}{36}$$
$$\frac{E_1 - E_2}{E_2 - E_3} = \frac{3}{4} \times \frac{36}{5} = \frac{27}{5}$$

81) In 2 years (730 days) around three Half life has been passed so after three half life remaining value of gadolinium will be 12.5%.

82)

$g_{eq.}$ of $N_2H_4 = g_{eq.}$ of BrO_3^-
moles × (V.F. = 4) = $\frac{2}{3}$ mole × (V.F. = 6)
Mole of $N_2H_4 = 1$

83) (ii) Let $r_1 \propto [H^+] \Rightarrow r_1 \propto [10^{-3}]^x$
[when pH = 3]
$r_2 = 100\ r_1 \propto [10^{-2}]^x$
[when pH = 2]
Thus, $\frac{r_2}{r_1} = \left[\frac{10^{-2}}{10^{-3}}\right]^x$

84) $NH_4HS_{(s)} \rightleftharpoons NH_{3(g)} + H_2S_{(g)}$
Eq. $\frac{P}{2}$ $\frac{P}{2}$
$K_p = \frac{P^2}{4} = \frac{(20)^2}{4} = 100$
$K_p = K_c \times (RT)^{\Delta n}$
$100 = K_c \times [0.0821 \times 400]^2$
$K_c = \frac{100}{1078.4} = 0.092M^2$

85)

Raoult's law is special case of Henry's law only ideal solutions obey Raoult's law.

86) $W_{Cu} = \frac{M_{Cu}}{2 \times F} It$

$\frac{W_{Cu}}{M_{Cu}} = \frac{It}{2F}$

$0.25 = \frac{I \times 193 \times 60}{2 \times 96500}$

I = 4.166 A current required when efficiency is 100%.

For 60% current efficiency = $\frac{4.166 \times 100}{60}$

I = 6.944 A.

87) $[I^-]$ from $BaI_2 = 2 \times 0.05 = 0.1$ M

$K_{(sp)AgI} = [Ag^+] \times [I^-]$

In presence of BaI_2,

$K_{sp} = (s = 10^{-15}) \times (0.1) = 10^{-16}$

In pure water,

$K_{sp} = s^2 = 10^{-16}$; $s = 10^{-8}$ M

88) $\Delta G = -T\Delta S_{universe}$

$\Delta S_{universe} = -\frac{\Delta G}{T} = -\frac{-11.5 \times 10^3}{700} = 16.43 JK^{-1}$

89) Statement-I is correct but Statement-II is incorrect.

90)

(HCOOH + HCOONa) = Acidic buffer

$pH = (pK_a = 14 - pK_b) + \log \frac{[Salt]}{[Acid]}$

 [Salt] = [Acid]

$pH = pK_a = 3.75$

BIOLOGY

91) NCERT-XI Pg. # 73

92) NCERT-XI, Pg.# 76.

93)

NCERT-XI, Pg. # 66

94)

NCERT-XI, Pg. # 62, 63

95) NCERT-XII, Page No. - XI, 65

96) NCERT-XII, Pg.#18.

97) NCERT-XII, Pg.# 9.

98) Module

99)

Allen module.

100)

NCERT-XI, Pg. # 16

101) NCERT Pg.# 32

102) NCERT-XI, Pg. # 17, 18

103) NCERT XI Pg. No. # 33, 32

104) NCERT-XI, Pg. # 27, 32

105) NCERT XI Page No.: 33

106) NCERT-XI, Pg. 29

107)

NCERT-XI, Pg. # 110

108) NCERT-XI, Page # 109

109) NCERT XI Pg # 112, 113, 114

110) NCERT-XI, Pg. # 137, 138

111) NCERT-XI, Pg#137,138

112) NCERT-XI, Pg. No. # 154

113) NCERT-XI, Pg. No. # 155, 156, 158, 160

114) NCERT XI Pg. # 157

115) NCERT-XI, Pg. # 177

116) NCERT-XI, Pg#175, 176

117) NCERT Pg.#55,58 (E), 62,65 (H)

118) NCERT Pg.#73-75

119) NCERT Pg.#69

120) NCERT-XII, Pg. # 73

121) NCERT-Pg. # 69

122) NCERT, Pg. # 81

123) NCERT, Pg. # 93-94

124) NCERT, Pg. # 92

125)

NCERT-Pg. # 96

126)

NCERT-XII, Pg. # 96

127) NCERT, Pg. # 93 (E), 103 (H)

128)

NCERT-XII, Pg. # 199

129) NCERT XII Pg. # 198, 199

130) NCERT Pg. # 197

131)

NCERT-XII, Pg. # 209

132) NCERT 12th, Page No. # 213

133)

NCERT-XII, Pg. # 219

134) NCERT XII Pg. No. # 224

135) NCERT Pg. # 224

136) NCERT, Pg. # 41

137) NCERT XIth Page # 52

138) NCERT XI Page # 53

139) NCERT XIth Page # 47

140) NCERT Page-111

141) NCERT page - 118 and 119

142) NCERT Page No. 101

143) NCERT_Page No.335

144) NCERT_Page No.338

145) NCERT_Page No. 337, 338

146) NCERT Page No # 321

147) NCERT-XI, Pg. No. # 317

148) NCERT-XI Page No. 223

149) NCERT Pg. # 217, 218

150) NCERT Pg. # 203,190

151)

NCERT, Pg. # 193, 194

152) NCERT Pg. # 202

153)

NCERT, Pg. # 212

154)

NCERT, Pg. # 205

155) NCERT Pg. # 190

156) NCERT Pg # 53

157)

NCERT, Pg. # 28

158) NCERT Pg # 58

159)

NCERT, Pg. # 38

160) NCERT Pg. # 32

161) NCERT-XII Pg#60,61

162) NCERT Pg. No. # 146-149, 153

163) NCERT Pg. No. # 152

164) NCERT Pg.# 138-140

165) NCERT-XII, Pg. # 111

166)

NCERT, Pg. 124, 125

167) NCERT, Pg # 123,124,126,127

168) NCERT Pg. No. # 126, 127

169)

NCERT-XI, Pg. # 121-124

170)

NCERT, Pg. # 100

171) NCERT Pg # 91, 97, 99, 100

172)

NCERT-XI, Pg. # 94, 97, 98, 100

173) NCERT Pg. # 170 (E) / 186 (H)

174) NCERT Pg. # 199

175)

NCERT, Pg. # 169

176) NCERT Pg. # 184

177)

NCERT, Pg. # 180

178)

NCERT-XII, Pg. # 179

179) NCERT-XII, Pg. # 153

180) NCERT XII, Pg. # 151